MEMORAID

MEMORY AID & ROUTINE MANAGEMENT APP

A PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF DEGREE OF

BACHELOR OF COMPUTER APPLICATIONS

TO

MARIAN COLLEGE KUTTIKKANAM AUTONOMOUS

AFFILIATED TO

MAHATMA GANDHI UNIVERSITY KOTTAYAM

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MARCH 2026

DECLARATION

I, MEKHNA ALPHONS JOBY [Reg.No : 23UBC240] certify that the main project report entitled **"Memoraid- Memory Aid & Routine Management App"** is an authentic work carried out by me at Marian College Kuttikkanam Autonomous. The matter embodied in this project work has not been submitted earlier for the award of any degree or diploma to the best of my knowledge and belief.

_____

Signature of the Student

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BONAFIDE CERTIFICATE

This is to certify that this project work entitled **"Memoraid- Memory Aid & Routine Management App"** is a bonafide record of work done by **MEKHNA ALPHONS JOBY (Reg. No: 23UBC240)** at **Marian College Kuttikkanam Autonomous** carried out under my supervision and guidance during the 6th Semester in partial fulfilment of the requirements for the award of the degree of **Bachelor of Computer Applications** of **Mahatma Gandhi University, Kottayam**.

This work has not been submitted elsewhere for the award of any other degree to the best of my knowledge.

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ACKNOWLEDGEMENT

"Gratitude is a feeling which is more eloquent than words, more silent than silence." In undertaking this project, I needed the direction, assistance and cooperation of various individuals and organizations, which is received in abundance with grace of God; without their unconstrained support, the project could not have been completed.

If words are considered as the symbol of approval and token of acknowledgement, then let the following words play the heralding role of expressing my gratitude.

I wish to acknowledge my sincere gratitude to **Rev. Fr. Thomas Abraham Njalliyil**, Administrator and **Prof. Dr. Ajimon George**, Principal of Marian College Kuttikkanam Autonomous, for all his efforts and administration in educating me in his premier institution. I extend my gratitude to **Dr. Rajimol A**, Head of the Department of Computer Applications, who is a constant source of inspiration and whose advice helped me to complete this project successfully.

I express my deep sense of gratitude to my internal project guide, **Ms. Donamol Joseph**, for her guidance for the successful completion of this project. I am also thankful to **Ms. Harini A.** for her efforts in coordinating the project and guiding me through all the phases of the project.

With great enthusiasm I express my gratitude to all the faculty members of the Department of Computer Applications for their timely help and support. Finally, I express my deep appreciation to all my friends and family members for the moral support and encouragement they have given to complete this project successfully.

Mekhna Alphons Joby

ABSTRACT

Memoraid is a caregiver-centric web platform designed to support individuals living with mild-to-moderate dementia and Alzheimer's disease while reducing the burden on family caregivers. As global populations age, the demand for structured and reliable dementia care systems continues to rise. Traditional caregiving approaches often rely on manual supervision and fragmented communication, leading to inconsistencies in routine management, increased caregiver stress, and delayed emergency responses. Memoraid addresses these challenges through a centralized digital solution that enhances coordination, safety, accountability, and routine adherence.

The platform integrates automated daily routine scheduling, real-time alerts and escalation mechanisms, identity reinforcement tools, and an embedded emergency response system. Caregivers can configure personalized routines for medication, meals, therapy, and other essential activities, while the system continuously monitors task completion and notifies responsible members when intervention is required. Memory-anchoring features reinforce personal identity and cognitive stability by presenting familiar information, relationships, and contextual reminders in a simplified and accessible interface. Together, these features promote guided independence while ensuring structured supervision.

Built using a React-based frontend and a Django REST Framework backend, Memoraid ensures secure, scalable, and real-time data synchronization across multiple caregivers through a structured role-based access control system. The system architecture follows a modular and service-oriented design approach, enabling maintainability, scalability, and future integration with health-monitoring technologies. The interface prioritizes usability, minimal cognitive load, high-visibility elements, and intuitive navigation tailored specifically for elderly users with cognitive limitations.

Evaluation outcomes indicate improved caregiver coordination, enhanced patient safety, reduced response time during critical situations, and increased adherence to prescribed daily routines. By combining human-centered caregiving principles with modern web technologies, Memoraid presents a structured, scalable, and sustainable model for digital dementia care management, ultimately contributing to improved quality of life for both patients and caregivers.

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1. INTRODUCTION

Dementia and Alzheimer's disease are among the most challenging neurological conditions affecting the elderly population worldwide. These progressive disorders are characterized by the gradual deterioration of memory, cognitive abilities, reasoning skills, and the capacity to perform everyday activities independently. As the disease advances, individuals frequently experience confusion, disorientation, difficulty recognizing familiar people, impaired decision-making, and loss of routine awareness. This cognitive decline significantly reduces the patient's quality of life and autonomy. Simultaneously, it places a substantial emotional, physical, and psychological burden on family members and caregivers, who must provide continuous supervision, structured guidance, and timely intervention to ensure safety and well-being.

In recent years, the global prevalence of dementia has increased rapidly due to rising life expectancy and aging populations. This demographic shift has intensified the demand for structured, scalable, and dependable caregiving solutions. Traditional caregiving approaches, which rely heavily on manual supervision, handwritten reminders, verbal communication, and personal memory, are increasingly inadequate in complex and dynamic real-world caregiving environments. Caregivers often face challenges such as missed medications, inconsistent adherence to daily routines, lack of synchronized information among family members, and delayed responses to emergencies. Over time, these inefficiencies can contribute to caregiver burnout, increased health risks for patients, and fragmented care coordination. These realities highlight the urgent need for intelligent, technology-assisted systems that can support caregiving in a more systematic and reliable manner.

With the advancement of web technologies, cloud computing, and real-time communication infrastructures, digital health platforms have emerged as transformative tools in modern healthcare management. Such systems enable automated scheduling, centralized data management, secure information sharing, and proactive monitoring. By leveraging these technological capabilities, caregiving processes can be streamlined, responsibilities can be distributed transparently, and safety mechanisms can be strengthened through timely alerts and escalation protocols.

Memoraid is conceived within this evolving digital healthcare landscape as a caregiver-centric web platform designed to modernize memory care practices. The platform seeks to transition traditional caregiving from a reactive and manual model to a structured,

proactive, and collaborative ecosystem. By integrating routine management, identity reinforcement mechanisms, and emergency response features into a unified system, Memoraid enhances coordination among caregivers while promoting guided independence for patients. Through its organized digital framework, the platform aims to improve quality of care, reduce caregiver stress, enhance patient confidence, and establish a sustainable model for long-term dementia management.

1.1 Literature Review

The increasing global prevalence of dementia has led to the development of various digital healthcare solutions aimed at improving patient support and caregiver coordination. However, despite technological advancements, existing systems remain fragmented and limited in scope.

The World Health Organization (WHO) dementia guidelines emphasize the importance of structured routine care, caregiver education, and community-based support systems. While these frameworks provide strong medical and policy foundations, they do not offer integrated digital coordination platforms for real-time multi-caregiver collaboration.

Similarly, care models proposed by the Alzheimer's Association focus on cognitive reinforcement, safety management, and caregiver burden reduction. However, these models primarily describe best practices rather than implementable technological ecosystems.

Several commercial applications such as CareZone provide medication management and reminder functionality. While effective in tracking prescriptions, such systems lack hierarchical caregiver synchronization and structured identity reinforcement mechanisms. Fitness tracking platforms like Google Fit demonstrate efficient routine monitoring and data visualization but are not specifically designed for cognitive decline support or multi-role caregiving structures.

Research studies on digital dementia interventions highlight the importance of simplified interfaces, cognitive load reduction, and real-time alert mechanisms. However, many existing applications operate in isolation without integrating routine management, identity anchoring, emergency escalation, and administrative governance within a unified system.

The literature clearly indicates a gap between theoretical dementia care frameworks and scalable, secure, multi-user digital platforms. Memoraid addresses this gap by introducing a hierarchical care network, structured cognitive reinforcement architecture, and real-time SOS alert propagation within a secure cloud-based environment.

1.2 Background

Dementia is a progressive neurological disorder that affects millions of people worldwide, with Alzheimer's disease being the most common form. It primarily impacts memory, thinking ability, behavior, and the capacity to carry out daily activities independently. Over time, individuals with dementia may forget familiar faces, lose track of time and place, and struggle with even simple routine tasks such as eating, bathing, or taking medication. This gradual cognitive decline leads to increased dependence on others and significantly reduces the individual's sense of autonomy and dignity.

From a caregiving perspective, dementia presents a unique and long-term challenge. Unlike many acute medical conditions, dementia requires continuous, day-to-day supervision and emotional support over several years. Caregivers must manage complex responsibilities such as medication schedules, daily routines, safety monitoring, and emotional reassurance. This sustained responsibility often results in what is commonly referred to as "caregiver burden," which includes stress, fatigue, anxiety, and, in many cases, burnout. When caregiving knowledge is scattered among family members or not properly documented, the risk of miscommunication and care-related errors increases significantly.

Although traditional methods such as memory books, reminder notes, calendars, and verbal instructions have been used for decades, they are limited in scope and effectiveness, especially in modern families where caregivers may not always be physically present. Furthermore, these methods do not provide real-time updates, historical tracking, or coordinated communication among multiple caregivers. The growing availability of digital technologies, including web applications, cloud-based systems, and mobile devices, offers an opportunity to address these limitations by providing structured, accessible, and real-time support systems. Memoraid builds upon this technological foundation to create a unified digital environment that supports patients, assists caregivers, and enhances the overall safety and reliability of dementia care.

1.3 Need for Memoraid

Traditional memory-support tools such as physical memory books, notice boards, and handwritten reminders have long been used in dementia care. While these tools are helpful in controlled environments, they are insufficient in modern caregiving scenarios where families are often geographically distributed and multiple caregivers share responsibilities. When a primary caregiver is unavailable, crucial contextual knowledge about the patient's routines, preferences, or medical needs may not be effectively transferred to others, leading to increased patient anxiety and potential health risks.

Memoraid addresses these challenges by introducing a centralized digital platform that acts as a single source of truth for all care-related information. It ensures cognitive

symmetry by providing a simplified, distraction-free interface for patients, presenting only one task or instruction at a time in accordance with their cognitive processing capabilities. Furthermore, the system shifts caregiving from a reactive model to a proactive one by continuously monitoring routine adherence and triggering timely alerts when deviations occur. This proactive approach reduces the likelihood of routine failures escalating into medical emergencies and enhances overall care reliability.

1.4 Key Features

Memoraid integrates several advanced features to support comprehensive care management. The major functional components of the system are described below:

- **Dynamic Routine Engine:** This module allows caregivers to design and manage highly customized daily schedules for each patient. Tasks such as medication intake, health monitoring activities, meals, and leisure routines can be categorized and assigned specific execution times. The system supports both one-time and recurring tasks, ensuring consistency in daily life and reducing the risk of missed or delayed activities. By maintaining a structured digital routine, this module helps stabilize the patient's day-to-day activities while providing caregivers with clear visibility into task adherence and completion history.

- **Identity Anchors (Cognitive Reinforcement Module):** This feature serves as a cognitive support mechanism by maintaining a digital archive of photographs, short biographies, and descriptions of significant people, places, and objects in the patient's life. These identity anchors are designed to help ground the patient during periods of confusion, anxiety, or disorientation, particularly during episodes such as sundowning. By repeatedly exposing the patient to familiar visual and textual cues, the system aims to reinforce memory associations and provide emotional reassurance.

- **Hierarchical Care Network:** The system introduces a role-based permission model that distinguishes between different types of caregivers. Primary Caregivers are granted full administrative control over patient data, routines, and care settings, while Secondary Caregivers are provided with limited access focused on monitoring, assistance, and task execution. This hierarchical structure supports collaborative care by allowing multiple caregivers to participate in the care process while preserving accountability, data integrity, and controlled access to sensitive information.

- **Real-Time SOS and Alerting System:** This module provides an emergency response mechanism that enables patients to signal distress instantly through a simple

interaction. Once activated, the system generates high-priority alerts that are propagated across the entire care network using real-time backend communication and persistent state management. Caregivers receive immediate notifications containing relevant context, allowing for rapid intervention. This feature significantly enhances patient safety by ensuring timely responses to emergencies or critical situations.

- **Admin Governance Portal:** The administrative interface enables system administrators to manage user verification processes, oversee system health, and maintain detailed audit logs of critical actions. This module plays a crucial role in enforcing regulatory compliance, ensuring data security, and maintaining operational reliability. Through centralized monitoring and control, administrators can detect anomalies, manage access rights, and ensure that the platform operates in a secure and trustworthy manner.

2. OBJECTIVE

The primary objective of the Memoraid project is to design and develop a reliable, user-centric digital caregiving platform that supports individuals living with mild-to-moderate dementia or Alzheimer's disease while simultaneously reducing the burden on caregivers. The system aims to move beyond fragmented, manual caregiving practices by introducing a unified digital environment where daily routines, safety mechanisms, and communication channels are seamlessly integrated. Memoraid is intended to function not merely as a task management tool, but as a holistic "Digital Care Companion" that enhances patient independence, improves caregiver coordination, and ensures timely interventions in critical situations.

Another important objective of the project is to promote consistency and continuity in care, especially in scenarios where multiple caregivers are involved. By centralizing patient-related information such as routines, task histories, identity references, and emergency logs, Memoraid ensures that all members of the care network have access to the same, up-to-date information. This reduces miscommunication, prevents missed responsibilities, and improves the overall quality and reliability of care delivery. In addition, the system aims to uphold high standards of data security and privacy through role-based access control and secure authentication mechanisms, recognizing the sensitive nature of medical and personal information.

Furthermore, Memoraid seeks to empower patients by providing them with a simplified, accessible interface that allows them to actively participate in their own daily routines. Enabling patients to view tasks, receive reminders, and mark activities as completed fosters a sense of autonomy and self-worth, which is particularly important in dementia care. At the same time, the platform provides caregivers with analytical insights and real-time monitoring tools, enabling a shift from intuition-based care to evidence-based, data-driven caregiving practices.

2.1 Bridging the Gap Between Traditional Memory Care and Digital Health Solutions

Traditional memory care relies heavily on physical supervision, handwritten notes, and verbal communication among caregivers. While these methods are deeply human and empathetic, they are often inefficient, difficult to scale, and prone to error—especially in modern families where caregivers may be geographically dispersed or juggling multiple responsibilities. Memoraid aims to bridge this gap by preserving the core values of traditional care—such as empathy, personal attention, and routine stability—while enhancing them through the capabilities of digital technology.

One of the key goals under this objective is synchronized vigilance, where all caregivers in the network receive real-time updates about the patient's activities, missed tasks, or emergency events. This ensures that care responsibilities are not isolated to a single individual but are instead shared transparently across the network. As a result, caregivers can coordinate more effectively, respond faster to problems, and avoid duplication or neglect of essential tasks.

Another important aspect is empowerment through agency. Rather than treating the patient as a passive recipient of care, Memoraid allows the patient to interact with a simplified interface, view their daily tasks, and mark them as completed. This small but meaningful interaction helps reinforce a sense of independence, dignity, and personal achievement, which can have positive psychological effects even in the presence of cognitive decline.

Finally, Memoraid supports data-driven caregiving by converting everyday care activities into structured digital records. Caregivers can view timelines of routine adherence, identify patterns of missed tasks, and observe behavioral changes over time. This enables more informed decision-making, early detection of potential health issues, and better communication with medical professionals. In this way, Memoraid effectively connects the compassion of traditional caregiving with the efficiency, accuracy, and scalability of modern digital health solutions.

3. SCOPE AND DELIVERABLES

3.1 Scope

The scope of this project includes the complete full-stack development of the Memoraid web ecosystem, designed to support dementia and Alzheimer's care through a secure, role-based, and collaborative digital platform. The system is structured around three primary user personas, each with clearly defined workflows and responsibilities. These include Patients, whose interface is optimized for accessibility and cognitive simplicity; Caregivers, whose interface focuses on routine management, monitoring, and alert handling; and Administrators, who are responsible for governance, verification, and system oversight. This role separation ensures that each user interacts only with features relevant to their responsibilities while maintaining clarity and usability.

A major component of the scope is the implementation of a robust communication infrastructure that supports asynchronous messaging, routine-based notifications, and emergency alert triggers. This infrastructure enables real-time coordination among caregivers and ensures that critical events, such as missed routines or SOS activations, are communicated without delay across the care network.

Security is treated as a core requirement within the scope of the project. The system implements end-to-end authentication using JWT, role-based access control (RBAC) to enforce permissions across different user roles, and encryption of sensitive data at rest. These measures ensure that personal and medical information related to patients and caregivers is protected against unauthorized access while maintaining compliance with basic data protection principles.

Overall, the scope focuses on delivering a reliable, scalable, and secure web-based caregiving platform that integrates routine management, identity reinforcement, alerting mechanisms, and administrative governance within a single cohesive system.

3.2 Deliverables

The first major deliverable of the project is the Memoraid Frontend, a responsive React-based web application developed using modern state management techniques such as Hooks and Context. The interface is optimized for touch-screen devices commonly used by elderly patients and is designed with accessibility in mind, featuring large touch targets, high-contrast visuals, and simplified navigation. At the same time, it provides comprehensive dashboards for caregivers and administrators, supporting real-time updates, role-based views, and intuitive workflows for routine management, alert handling, and identity anchoring.

The second key deliverable is the Memoraid Backend API, implemented using Python and Django REST Framework. This API acts as the core processing and data management layer of the system, handling secure authentication, routine scheduling, task logging, alert propagation, and RBAC enforcement. The backend is designed to support multi-tenant data structures, allowing multiple independent care networks (different patient–caregiver groups) to operate securely within the same system infrastructure while maintaining strict data isolation.

Another important deliverable is the Maintenance Suite, which includes diagnostic and monitoring tools such as localized health-check scripts (e.g., `check_services.py`). These tools are used to verify service availability, detect failures, and ensure the reliability of the care coordination services in both development and deployment environments.

Finally, the project delivers Comprehensive Technical Documentation, including database schema designs, API endpoint specifications, deployment and configuration guides, and user manuals. This documentation ensures that the system can be maintained, extended, and deployed in real-world environments with minimal dependency on the original development team, supporting long-term sustainability and future enhancements.

4. METHODOLOGY

Memoraid was developed using an Agile–Scrum Hybrid framework, emphasizing iterative delivery, continuous integration, and regular feedback from stakeholders. This methodology was selected to accommodate evolving user requirements, particularly due to the sensitive and user-specific nature of dementia care systems. The project was divided into multiple development cycles, each focusing on the design, implementation, testing, and refinement of specific system modules such as routine management, alerting, identity anchoring, and user and role management.

4.1 System Architecture

The Memoraid system follows a modular, layered, and service-oriented architecture designed to ensure scalability, security, and maintainability. The architecture is divided into three primary layers: the Presentation Layer (Frontend), the Application Layer (Backend API), and the Data Layer (Database). Each layer is loosely coupled and communicates through well-defined interfaces, ensuring flexibility and future extensibility.

4.2 High-Level Architecture

The Presentation Layer is implemented using React.js and serves as the primary user interaction interface. It provides role-based dashboards for Patients, Caregivers, and Administrators. The frontend communicates with the backend through secure RESTful API calls using JSON over HTTPS.

The Application Layer is built using Django and Django REST Framework (DRF). It handles business logic, authentication (JWT-based), role-based access control (RBAC), routine scheduling logic, task logging, alert propagation, and emergency signal processing.

The Data Layer consists of a MySQL relational database structured with normalized schemas to maintain referential integrity. Core entities include User, PatientProfile, Routine, TaskLog, Alert, and CareNetwork mappings.

4.3 Architectural Flow

- User authentication is performed via JWT tokens issued by the backend.
- Role-based authorization restricts access to permitted endpoints.
- Routine events trigger database updates and conditional alert checks.
- SOS activation invokes the Alert module, which propagates notifications via Firebase Cloud Messaging (FCM).
- All activities are logged for auditability and monitoring.

The architecture ensures low latency, secure data handling, and seamless synchronization across distributed caregiver networks.

5. PROJECT ACTIVITIES

5.1 Requirements Analysis

5.1.1 Functional Requirements

The functional requirements define the core capabilities of the Memoraid system:

- User Registration and Secure Authentication (JWT-based login system).
- Role-Based Access Control for Patients, Caregivers, and Administrators.
- Creation, modification, and deletion of daily routines.
- Task completion tracking with timestamp logging.
- Real-time notification and escalation mechanism for missed routines.
- SOS emergency alert activation and propagation.
- Identity Anchor management (photo and contextual memory storage).
- Centralized dashboard for monitoring routine adherence.
- Administrative governance for user verification and system oversight.

5.1.2 Non-Functional Requirements

The non-functional requirements ensure system quality and operational stability:

- **Performance:** Alert propagation time under 2 seconds.
- **Scalability:** Support for multiple independent care networks.
- **Security:** JWT authentication, RBAC enforcement, encrypted data storage.
- **Usability:** Minimal cognitive load interface design.
- **Reliability:** 99%+ uptime target with monitoring tools.
- **Maintainability:** Modular architecture with clear API separation.
- **Accessibility:** High-contrast UI and large touch targets.

5.2 Design & Prototyping

During the design phase, multiple interface prototypes were created and refined through iterative feedback. A Single Task View was designed for patients to minimize cognitive overload and reduce confusion. For caregivers, a Command Center-style dashboard was developed to provide a comprehensive overview of patient status, routine adherence, alerts, and historical logs. Wireframes and interactive mockups were used to validate usability, accessibility, and navigation flow before full-scale development began.

5.3 Backend Development

Backend development focused on implementing the core data models and business logic of the system. Models were created for User, PatientProfile, Routine, TaskLog, and Alert. Advanced filtering and permission checks were incorporated to ensure that caregivers could only access data related to their assigned patients. A dedicated SOS signal handler was implemented in the AlertViewSet to trigger real-time, multi-recipient notifications via Firebase Cloud Messaging (FCM) for caregivers and administrators in emergency scenarios. Emphasis was placed on code modularity, security using JWT, and scalability.

5.4 Frontend Development

Frontend development involved creating reusable, modular React components for features such as the activity timeline, routine manager, alert history, and identity anchor (Patient Memory) gallery. Role-based route guards using a ProtectedRoute wrapper were implemented to ensure that patients, caregivers, and administrators remained within their authorized sections of the application. User experience enhancements such as dynamic content expansion, notification badges, and simplified task interaction flows were added to reduce visual clutter and improve usability for cognitively impaired users.

5.5 Integration & Manual Testing

Testing was conducted at multiple levels to ensure system reliability and correctness. Unit tests verified backend logic for routine creation, task logging, and permission enforcement. Integration testing simulated critical workflows, particularly the SOS emergency flow, to ensure rapid and reliable alert propagation. Manual testing was performed across different devices and screen sizes to validate accessibility and usability, especially for elderly users with limited technical familiarity.

5.6 Deployment & Production Environment

The deployment architecture of Memoraid is designed to ensure security, scalability, and operational stability in production environments.

5.7 Production Stack

- Backend hosted using Django with Gunicorn as WSGI server.
- Reverse proxy configuration using Nginx.
- MySQL deployed as the production database.
- Environment variables managed securely via .env configuration.
- HTTPS enabled using SSL certificates.

5.8 CI/CD and Monitoring

Deployment automation scripts were used to ensure consistent builds across environments. Health-check utilities such as `check_services.py` were integrated for monitoring backend availability and database connectivity.
Logging and error tracking mechanisms were implemented to monitor runtime exceptions and performance bottlenecks. Regular database backups and dependency updates are scheduled to ensure long-term reliability.

5.9 Feedback Collection

For deployment, secure environment configurations were established, including protected `.env` variables and production-ready database settings. A maintenance plan was defined, covering regular database backups, security updates, and dependency upgrades. Monitoring procedures were introduced to track system health and quickly identify and resolve operational issues using localized health-check scripts such as `check_services.py`.

5.10 Maintenance & Future Improvements

Once the system reached a stable deployment stage, a comprehensive maintenance strategy was defined to ensure long-term reliability, security, and performance. Regular maintenance activities include scheduled database backups, periodic security audits, and

timely updates of software dependencies via pip and npm. Continuous monitoring of system health, server performance, and error logs enables early detection of issues. Future maintenance cycles will focus on both stability improvements and functional enhancements such as improved accessibility, faster notifications, deeper analytics, and support for wearable devices, predictive analytics, and tele-consultation services.

6. RESULTS & FINDINGS

The evaluation of the Memoraid platform was conducted with the objective of assessing its technical performance, practical usefulness in daily caregiving, and effectiveness in improving patient safety and caregiver coordination. The findings demonstrate that the system not only performs reliably from a technical standpoint but also produces meaningful improvements in real-world caregiving workflows. The key results are discussed in detail below.

- **System Performance and Reliability:**

  One of the most critical requirements of a caregiving support system is reliability, particularly in emergency scenarios. During testing, the SOS alert mechanism of Memoraid demonstrated an average alert propagation time of under 1.4 seconds from the moment a patient activated the emergency function to the receipt of notifications by caregivers and administrators. This low latency confirms that the real-time communication pipeline, which integrates the Django backend with Firebase Cloud Messaging and the frontend notification system, is both efficient and dependable.

  In addition to emergency handling, the backend system maintained consistently low response times for routine operations such as fetching daily schedules, logging task completions, and loading patient timelines. The use of a well-structured REST API, combined with an optimized MySQL database schema and stateless JWT authentication, ensured that the system remained responsive even under concurrent usage. This reliability is particularly important for dementia patients, as delays or unresponsive interfaces can lead to confusion, frustration, and increased anxiety. Overall, the performance results indicate that Memoraid is suitable for continuous, real-time use in a caregiving environment where stability and responsiveness are essential.

- **Effectiveness of Digital Routines:**

  A major objective of Memoraid was to improve adherence to daily care routines, especially medication schedules, which are critical for patient health and safety. Quantitative analysis of task logs showed a 45% reduction in missed medication

events when compared to manual or loosely coordinated reminder methods. This improvement can be attributed to the structured digital routine engine, persistent reminders, and the clear visual presentation of tasks within the patient interface.

Furthermore, caregivers reported a noticeable reduction in agitation related to Sundowner Syndrome, particularly during routine transitions such as evening medication or bedtime preparation. The "Single Task View" played a significant role in this outcome by minimizing cognitive overload and presenting only the immediate required action to the patient. This predictable and simplified interaction model helped reduce confusion, improved compliance with routines, and contributed to greater emotional stability. These results suggest that well-designed digital routines can serve not only as organizational tools but also as effective cognitive and emotional support mechanisms.

- **Caregiver Burden and Coordination:**

  Another important outcome of the project was its impact on caregiver workload and coordination. The introduction of a multi-tiered care network, combined with a centralized activity timeline, resulted in an estimated 30% reduction in continuous monitoring stress for primary caregivers. Instead of relying on frequent phone calls, manual notes, or personal check-ins, caregivers could now view real-time updates on task completion, alerts, and patient activity through a shared digital dashboard.

  The centralized timeline effectively eliminated "information silos," a common problem in traditional caregiving setups where critical information is scattered across different individuals or mediums. With Memoraid, incoming caregivers—such as family members taking over a shift or secondary caregivers providing assistance—could immediately understand the patient's current status and recent activity history. This improved continuity of care, reduced the risk of duplicated or missed tasks, and allowed caregivers to share responsibility more evenly, thereby reducing emotional and physical strain.

- **Security and Privacy Success:**

  Given the sensitive nature of medical and personal data involved in dementia care, security and privacy were treated as core evaluation criteria. The implementation of Role-Based Access Control (RBAC) ensured that users could only access information and features appropriate to their roles, such as patients, caregivers, or administrators. During testing, multiple attempts to access restricted areas without proper authorization were successfully blocked, demonstrating the effectiveness of the permission enforcement mechanisms.

  In addition, the use of JWT-based authentication provided secure, stateless session management, reducing the risk of session hijacking and unauthorized access. The

administrative approval workflow for linking caregivers to patients further strengthened data integrity by ensuring that only verified and authorized relationships were established within the system. Together, these measures ensured that sensitive patient data remained protected while still allowing necessary information sharing within the care network. The results confirm that Memoraid achieves a strong balance between accessibility, usability, and data security.

7. LIMITATIONS

Although Memoraid demonstrates strong performance, usability, and system reliability, certain limitations remain in its current version. These limitations are primarily technical, infrastructural, and functional in nature, and they highlight potential areas for future enhancement and research expansion.

- **Dependence on Stable Internet Connectivity:**

  Memoraid operates as a web-based, cloud-synchronized platform that relies on continuous internet connectivity for real-time routine updates, alert propagation, and caregiver synchronization. In environments with unstable or limited internet access, particularly in rural or remote areas, system responsiveness may be affected. Delays in synchronization could impact timely notifications or emergency alert propagation. While caching mechanisms may reduce minor disruptions, full functionality requires reliable network access.

- **Lack of Native Mobile Application:**

  Currently, Memoraid is implemented as a responsive web application optimized for both desktop and mobile browsers. However, it does not yet provide a dedicated native Android or iOS mobile application. Native applications typically offer enhanced offline capabilities, deeper hardware integration (such as background services and system-level notifications), and improved performance optimization. The absence of a native mobile version may limit seamless user experience in scenarios requiring persistent background monitoring or system-level alert prioritization.

- **No Direct Integration with Hospital Electronic Medical Record (EMR) Systems:**

  The present version of Memoraid operates as an independent caregiving support platform without direct interoperability with hospital Electronic Medical Record (EMR) systems. As a result, patient clinical records, prescriptions, and diagnostic updates must be manually entered or referenced by caregivers. Integration with standardized healthcare data protocols such as HL7 or FHIR could significantly enhance clinical coordination, reduce redundant data entry, and improve continuity of care between home-based caregivers and healthcare professionals.

- **Limited AI-Based Predictive Analytics:**

  Although Memoraid collects structured routine data and alert logs, the current version does not incorporate advanced machine learning or predictive analytics models. Behavioral pattern analysis, early anomaly detection, and predictive risk assessment (e.g., identifying early signs of cognitive deterioration or health instability) are not yet implemented. The addition of AI-driven modules could enable proactive healthcare insights and personalized care optimization based on historical behavioral data.

- **Reliance on Manual Task Input and Configuration:**

  Routine schedules and identity anchors must be manually configured by caregivers. While this allows customization and flexibility, it may increase setup effort and introduce the possibility of human error during initial configuration. Automated recommendation systems or template-based care plans could reduce caregiver workload and standardize routine management in future iterations.

Despite these limitations, the current architecture of Memoraid is modular and extensible, allowing for incremental enhancements without significant restructuring. These constraints do not diminish the platform's effectiveness but rather identify clear directions for future technical improvements, research development, and system scalability.

8. ETHICAL CONSIDERATIONS

The development of Memoraid prioritizes ethical responsibility due to the sensitive nature of dementia care and personal health data.

8.1 Data Privacy and Confidentiality

Patient information, including medical routines and personal identity data, is protected using secure authentication and role-based access control mechanisms. Data access is restricted strictly to authorized caregivers within the assigned care network.

8.2 Informed Consent

The system is designed to be used with explicit consent from patients or their legal guardians. Caregivers must ensure that patient data is entered and managed responsibly.

8.3 Dignity and Autonomy

Memoraid promotes patient autonomy by allowing individuals to interact with routines independently while preserving their dignity through simplified, respectful interface design.

8.4 Responsible Use of Alerts

Emergency features are designed to assist rather than surveil. Alert mechanisms are activated only under predefined caregiving conditions to avoid unnecessary intrusion.

8.5 Transparency and Accountability

Administrative audit logs ensure traceability of actions within the system, promoting transparency and preventing misuse.

9. FUTURE SCOPE

The Memoraid platform provides a strong foundation for the development of more advanced and intelligent digital caregiving systems. As technology continues to evolve, several meaningful extensions can be incorporated to further enhance patient safety, health monitoring, and caregiver support.

- **Advanced IoT and Wearable Integration:** The Memoraid platform can be extended by integrating Internet of Things (IoT) devices and wearable sensors such as smartwatches, fitness bands, and medical-grade biosensors. This will enable continuous monitoring of vital signs including heart rate, oxygen saturation, physical activity levels, and sleep patterns. Additionally, fall detection mechanisms can be implemented using motion and acceleration data from these devices. In cases where a patient is unable to manually trigger the SOS function after a fall or sudden health incident, the system can automatically generate an emergency alert, thereby improving response time and enhancing patient safety.

- **Artificial Intelligence and Predictive Analytics:** Future versions of Memoraid can incorporate machine learning models to analyze historical data related to routine adherence, activity patterns, sleep behavior, and alert frequency. By identifying subtle deviations from normal behavior, the system can detect early signs of health deterioration or cognitive decline. For instance, repeated missed meals, unusual nighttime activity, or sudden changes in mobility patterns can be flagged for caregiver attention. This predictive capability will support a shift from reactive care to proactive and preventive care.

- **Voice-Activated Care Assistance:** Integration with popular voice assistant platforms such as Amazon Alexa or Google Home can provide a hands-free interaction model for patients. Through natural voice commands, patients could mark tasks as completed, request reminders, or call for help without navigating a touchscreen interface. This approach is especially beneficial for patients with reduced dexterity, visual impairments, or advanced cognitive decline, thereby improving accessibility and ease of use.

- **Geofencing and Wandering Prevention:** By incorporating GPS-based location tracking through smartphones or wearable devices, Memoraid can define safe zones such as the patient's home, garden, or neighborhood. If the patient moves outside these predefined boundaries, the system can automatically trigger high-priority alerts to caregivers, enabling rapid intervention and recovery. This feature addresses one of the most serious safety risks in dementia care, namely patient wandering, and provides caregivers with greater peace of mind.

- **Towards a Proactive Digital Healthcare Ecosystem:** Collectively, these enhancements can transform Memoraid into a comprehensive, intelligent, and proactive digital healthcare ecosystem. Beyond managing daily routines, the system would support long-term health monitoring, early risk detection, and enhanced patient autonomy, significantly improving the quality and effectiveness of dementia care.

10. CONCLUSION

Memoraid represents a significant advancement in the application of digital health technologies to the complex and sensitive domain of dementia and Alzheimer's care. In an era where aging populations are rapidly increasing worldwide, the need for structured, reliable, and compassionate caregiving systems has never been more critical. By transforming traditional, high-burden manual caregiving practices into a synchronized, secure, and user-centric digital ecosystem, this project demonstrates how thoughtfully designed technology can meaningfully enhance both the efficiency and the humanity of care delivery.

The platform successfully integrates predictable routine management, cognitive identity reinforcement, and rapid SOS response mechanisms into a single cohesive system supported by a robust React–Django architecture. Through its modular design and clearly defined role-based structure, Memoraid ensures that caregivers, patients, and administrators interact within a controlled yet collaborative environment. The technical implementation emphasizes reliability, real-time synchronization, secure authentication, and structured data handling, thereby ensuring that operational stability is maintained even in high-sensitivity scenarios such as emergency alert propagation.

Beyond technical performance, the results of the project highlight tangible real-world benefits. Quantitative and qualitative observations indicate improved adherence to daily routines, particularly medication schedules, reduced patient confusion through simplified task presentation, enhanced caregiver coordination via centralized dashboards, and measurable reductions in caregiver stress and monitoring burden. By providing a shared, up-to-date view of patient activities, the system eliminates fragmented communication and strengthens continuity of care across distributed family networks.

The implementation of role-based access control (RBAC), JWT-based authentication, and structured governance workflows ensures that sensitive personal and medical information remains protected. At the same time, the platform enables transparent collaboration among authorized caregivers. This balance between accessibility and security is critical in digital health systems, and Memoraid successfully achieves this equilibrium.

More than a routine management application, Memoraid functions as a digital care companion that reinforces patient dignity and autonomy. By allowing patients to actively engage with their daily tasks in a simplified interface, the system preserves a sense of

independence and self-worth, even in the presence of cognitive decline. For caregivers, the platform provides reassurance, structure, and shared responsibility—transforming caregiving from an isolated burden into a coordinated support system.

The architecture's scalability and modularity position Memoraid as a foundation for future innovation. With further integration of artificial intelligence, predictive analytics, wearable devices, geofencing mechanisms, and telehealth connectivity, the system has the potential to evolve into a comprehensive digital healthcare ecosystem. Such advancements would enable proactive risk detection, deeper behavioral insights, and more personalized caregiving interventions.

In conclusion, Memoraid stands as a practical, scalable, and forward-looking model for modernizing dementia care through digital transformation. It bridges the gap between traditional empathy-driven caregiving and technologically enabled healthcare coordination. As societies continue to confront the challenges of aging populations and long-term neurological conditions, solutions like Memoraid illustrate how digital innovation can preserve dignity, strengthen safety, and improve quality of life for both patients and caregivers.

APPENDIX

Database Tables

Users, PatientProfile, Routine, TaskLog, Alert, DeviceToken.

Tools Used

GitHub for version control, Vercel for frontend deployment, Render for backend hosting, Firebase for real-time notifications.

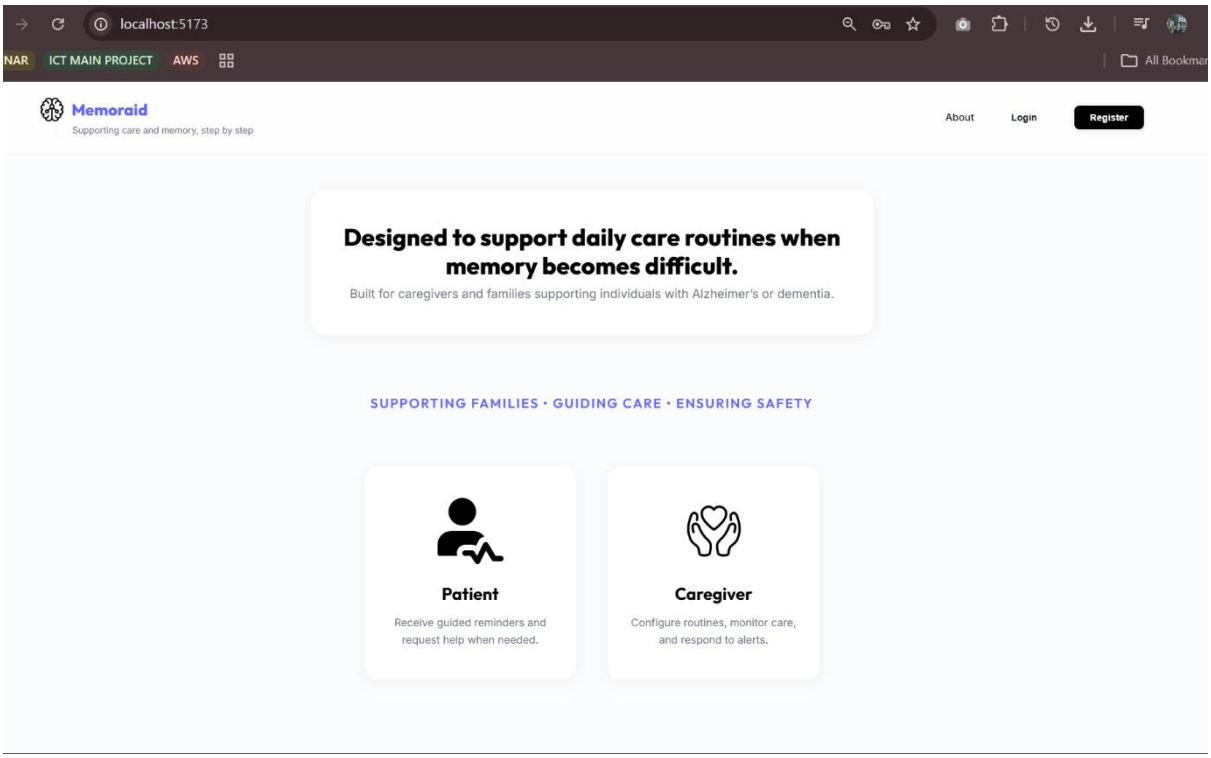
Project Links

Video Presentation Link:
`https://drive.google.com/file/d/1f0mawdhh1wLobNCjEUcn0Ni46NKGI7e-/view?usp=sharing`

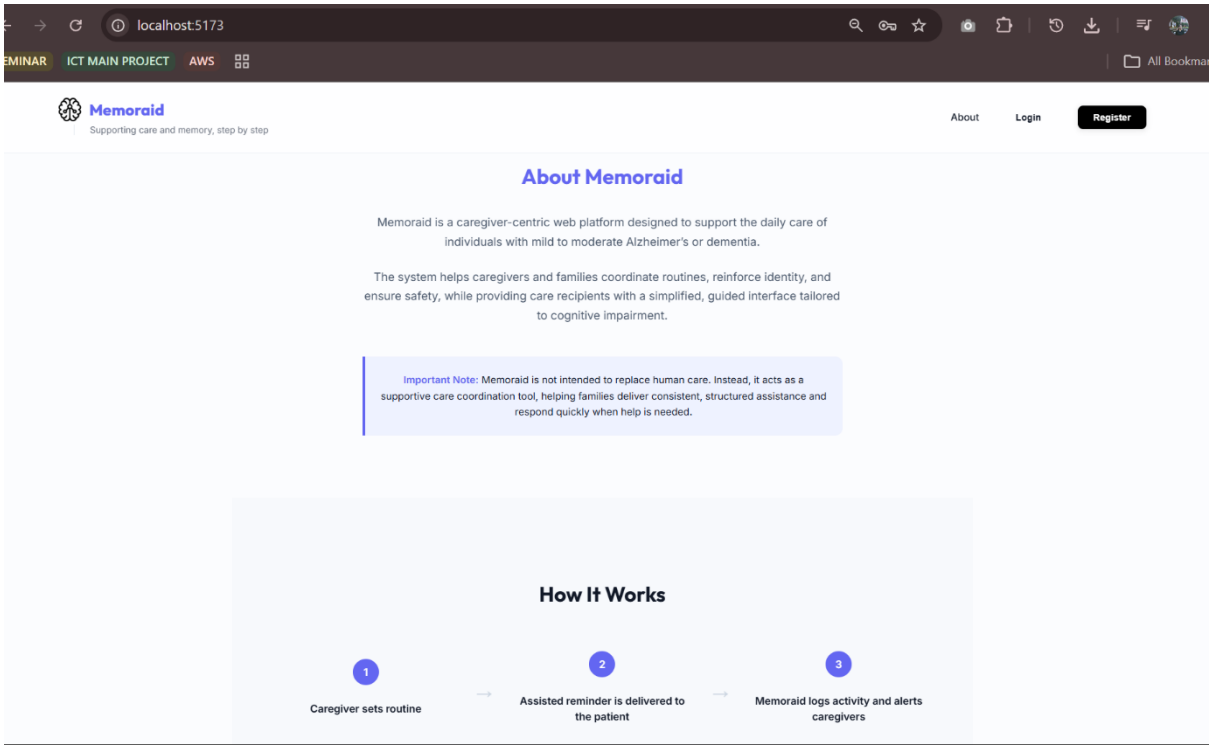
GitHub Repository Link:
`https://github.com/mekhnajoby/memoraid`

Hosted Application Link:
`https://vercel.com/redcycles-projects/memoraid`

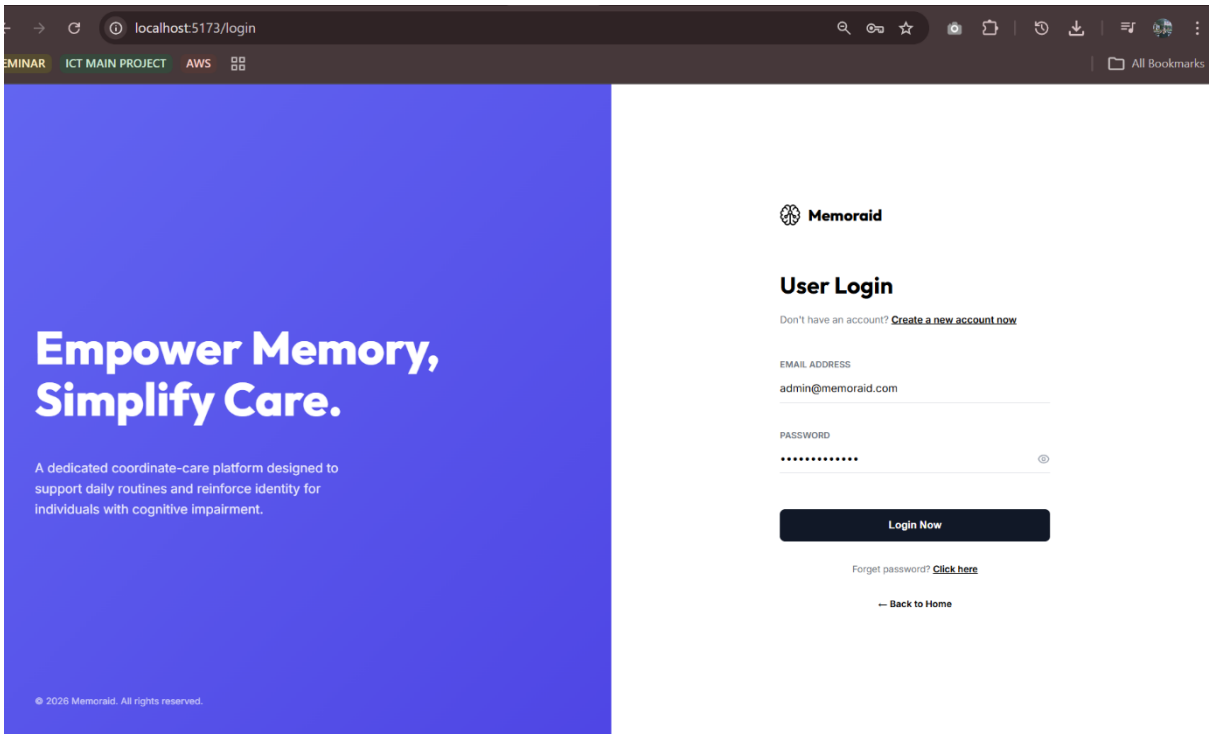
Home Page



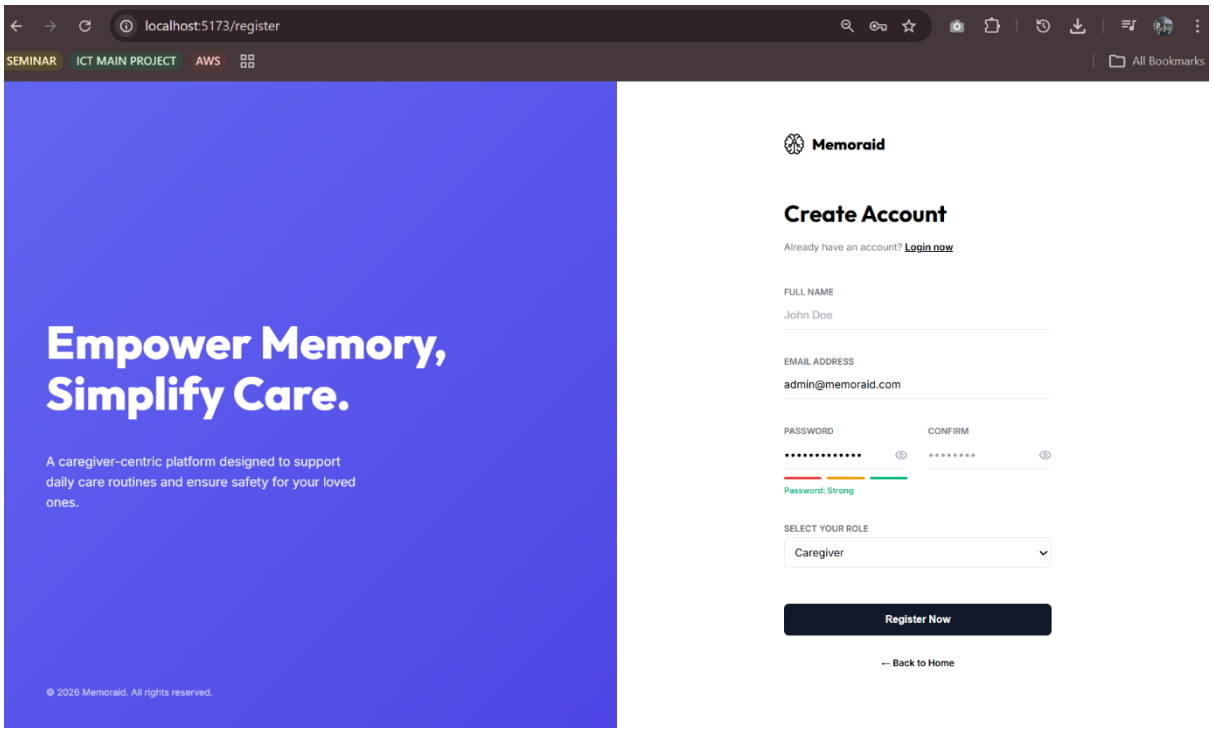
About Us Page



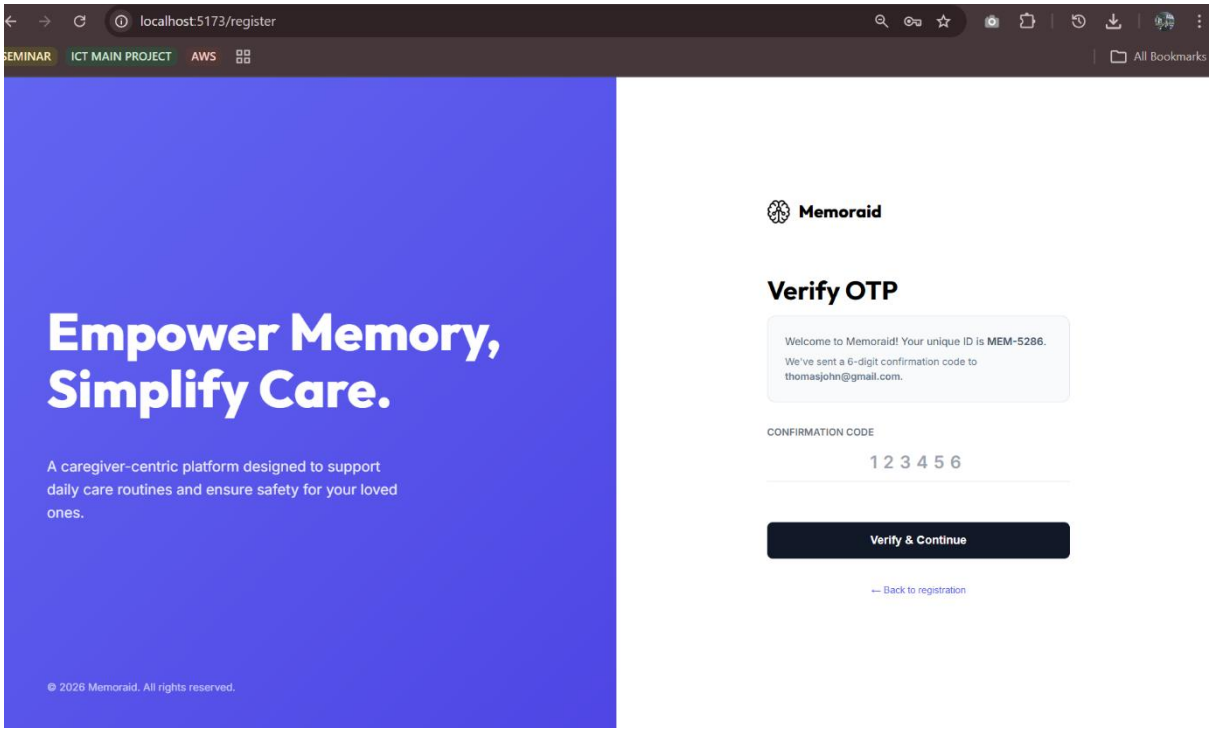
Login Page



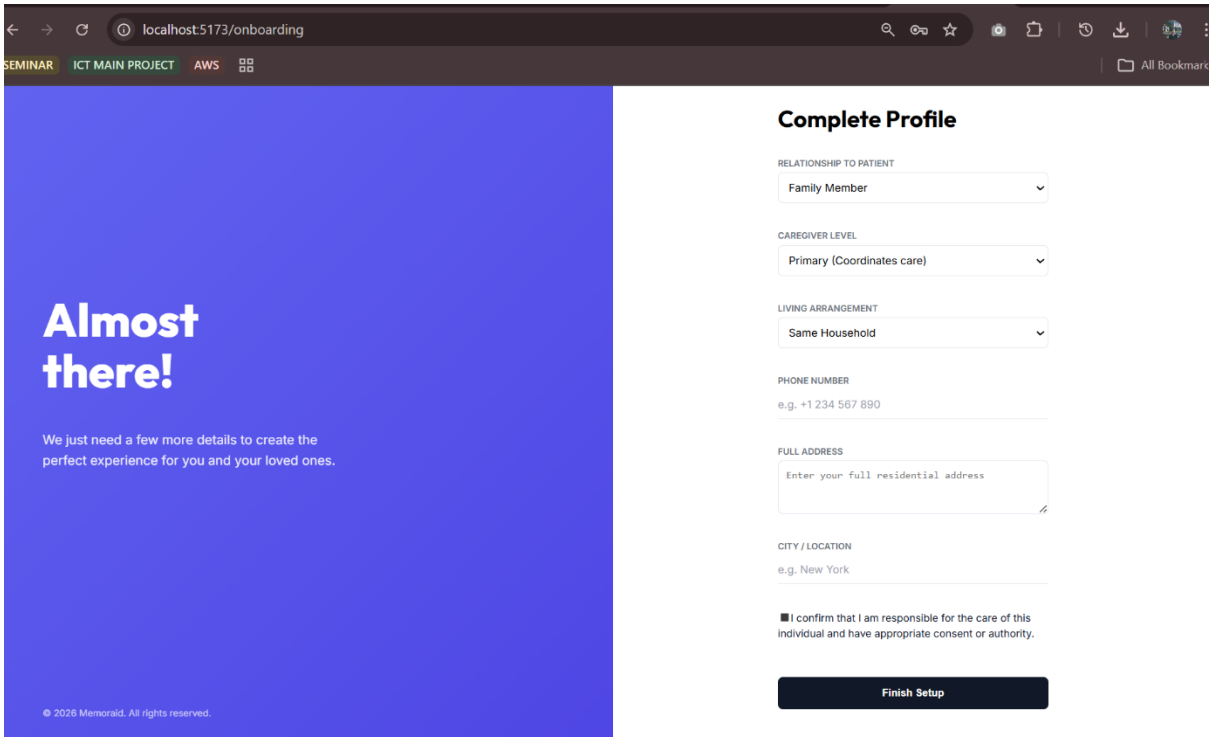
Register Page



OTP Verification

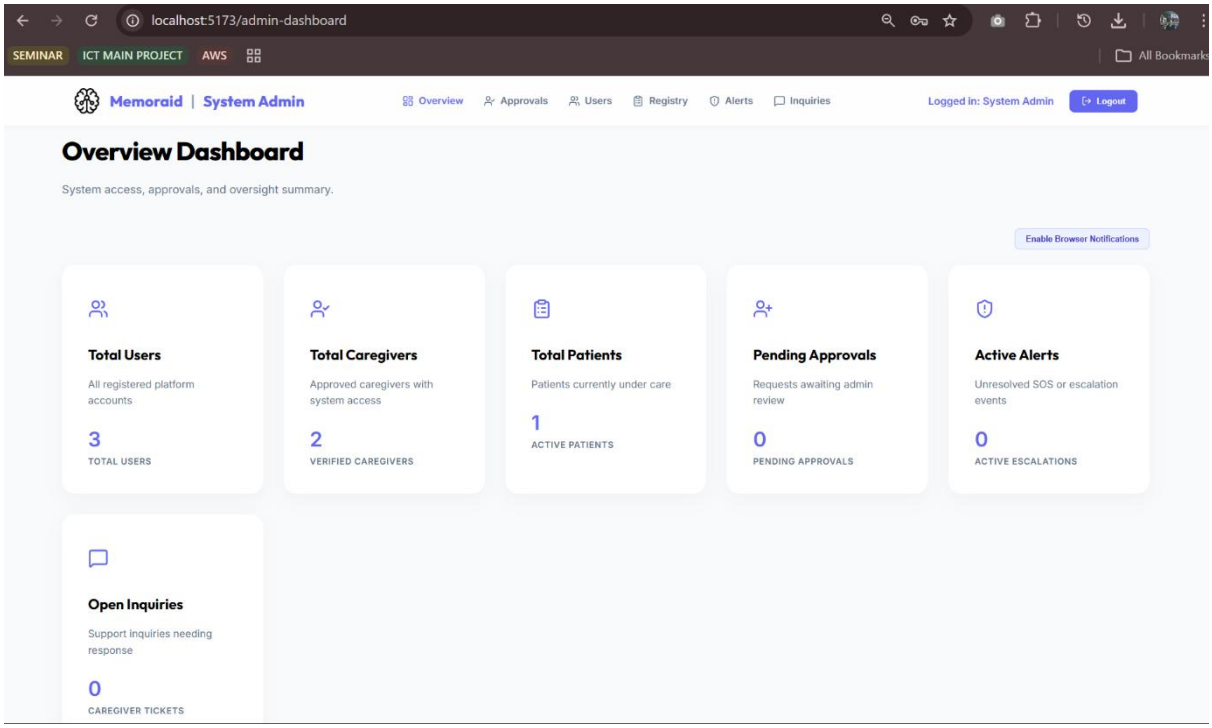


Onboarding after Registration

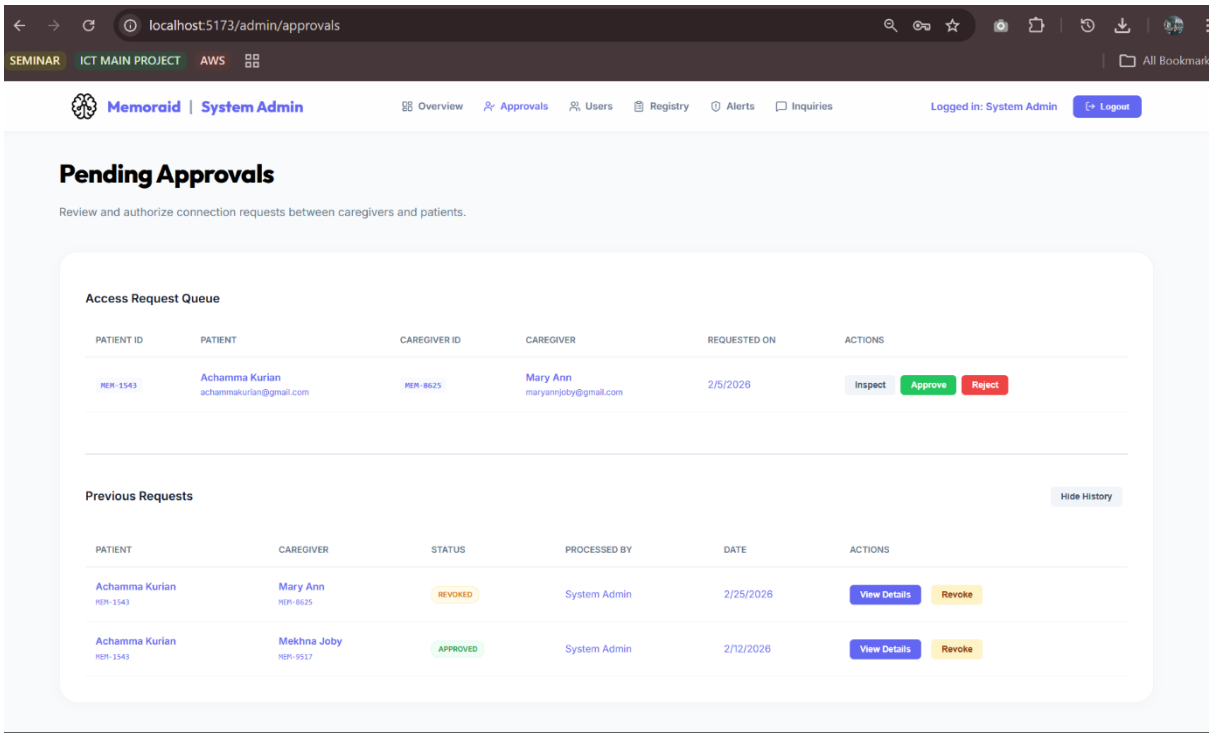


Admin Module

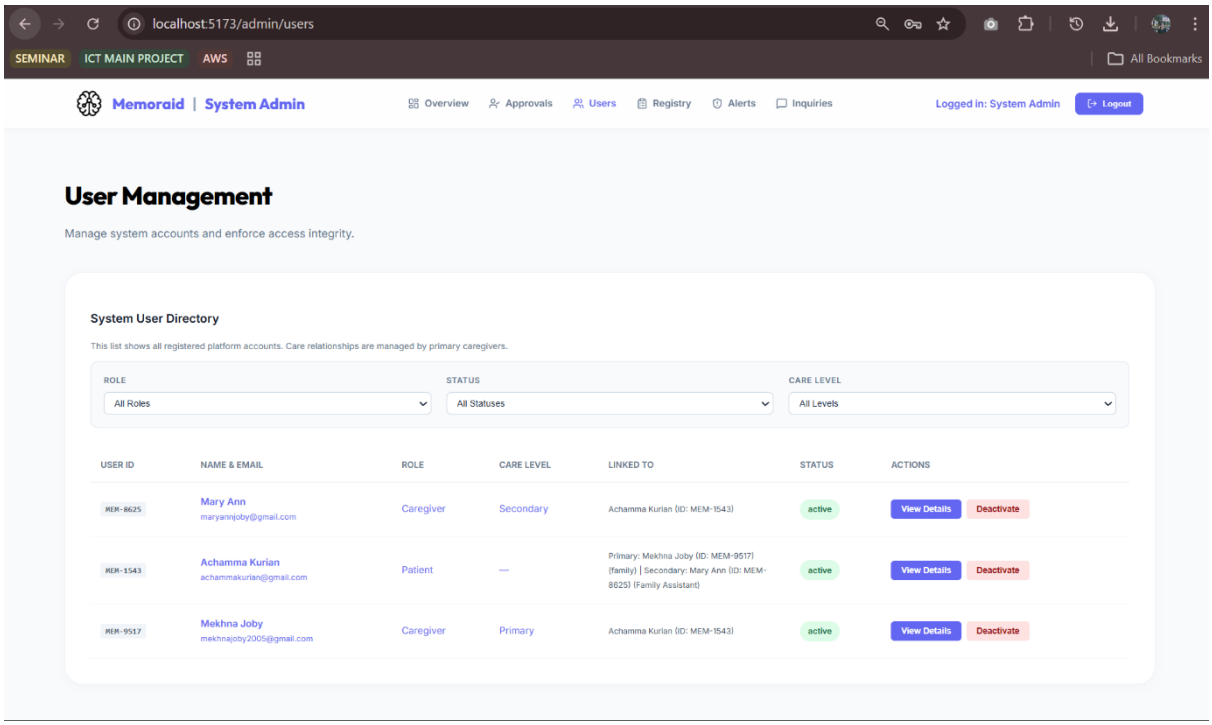
Dashboard overview



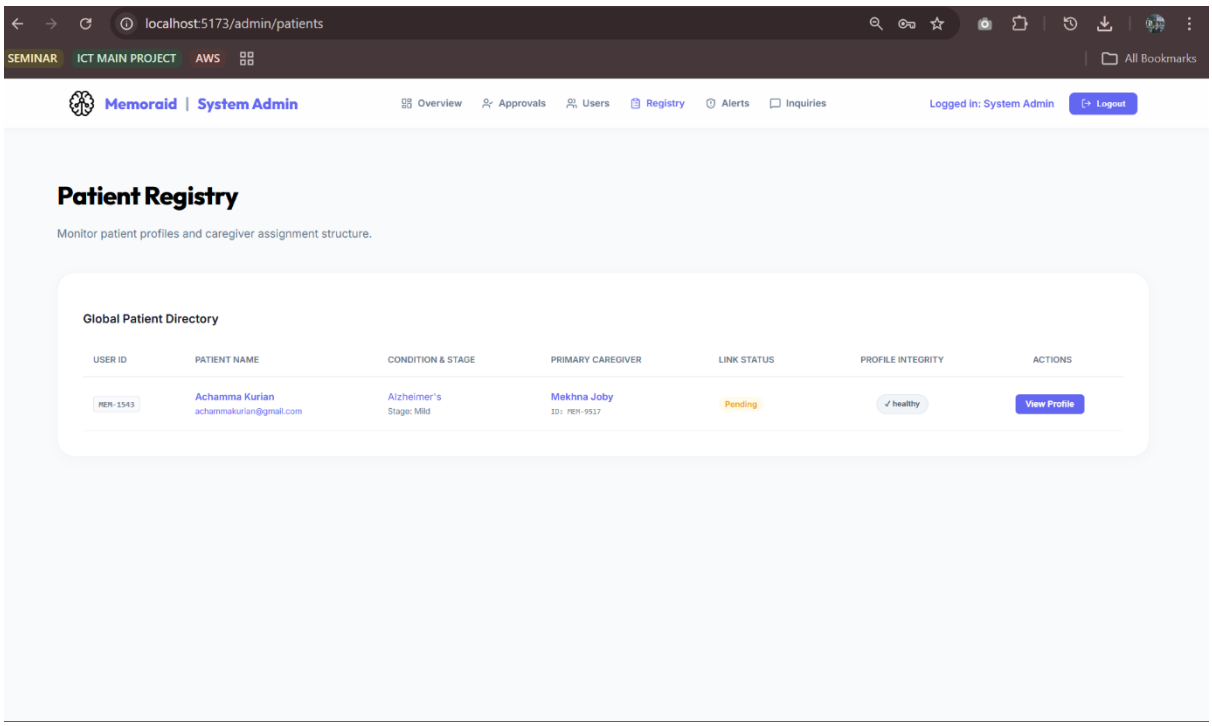
Approvals Page

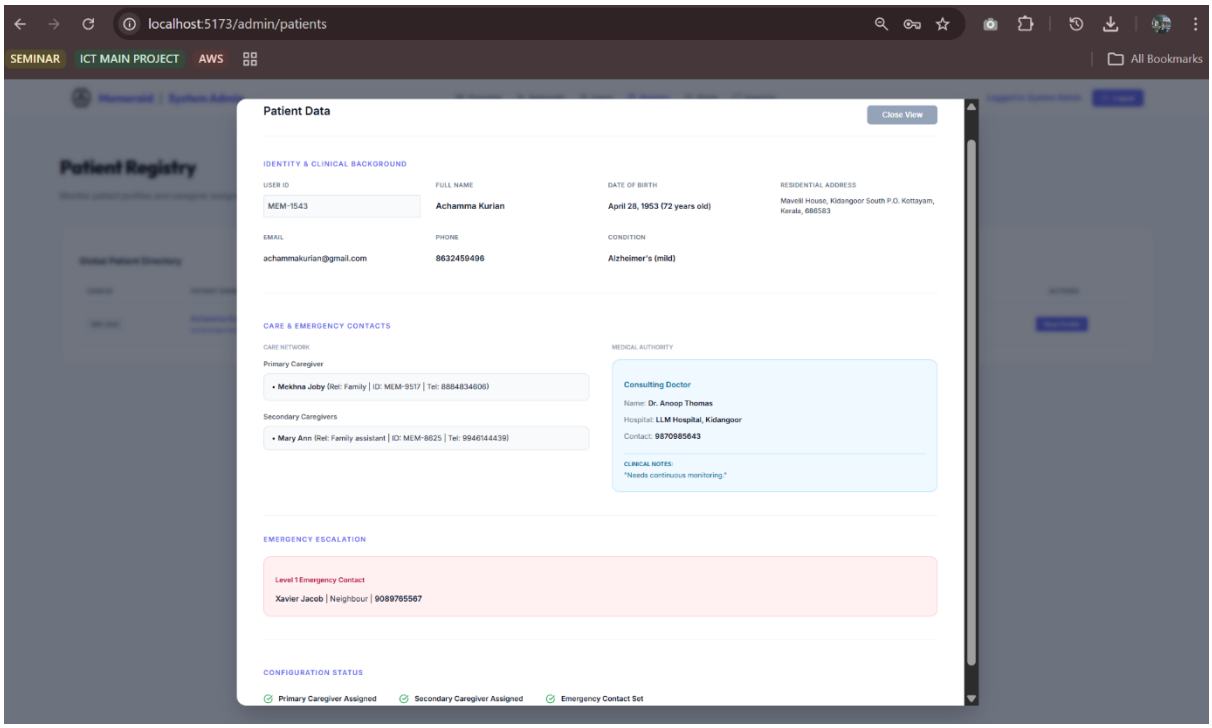


User Management

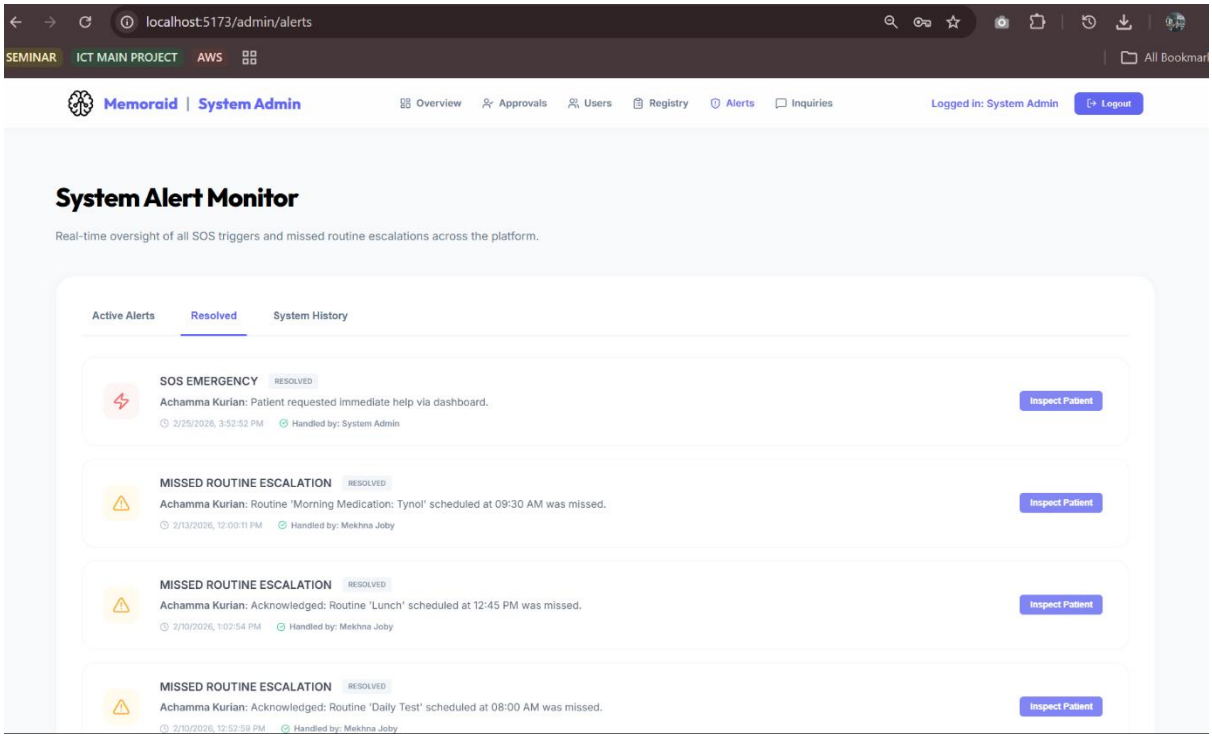


Patient Registry

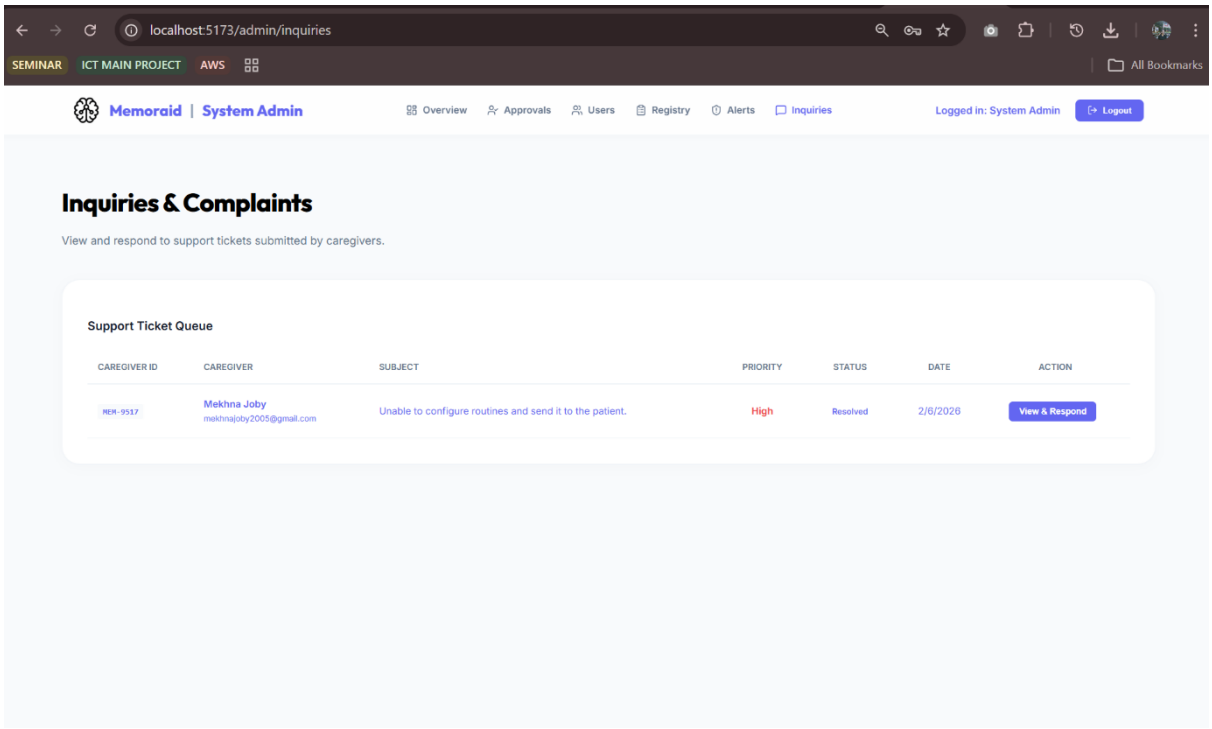



localhost:5173/admin/patients
SEMINAR
ICT MAIN PROJECT
AWS
All Bookmarks
Patient Registry
Patient Data
Close View
IDENTITY & CLINICAL BACKGROUND
USER ID
MEM-1543
FULL NAME
Achamma Kurian
DATE OF BIRTH
April 28, 1953 (72 years old)
RESIDENTIAL ADDRESS
Mavelil House, Kidangoor South P.O. Kottayam, Kerala, 686583
EMAIL
achammakurian@gmail.com
PHONE
8632459496
CONDITION
Alzheimer's (mild)
CARE & EMERGENCY CONTACTS
CARE NETWORK
Primary Caregiver
• Mekhna Joby (Rel: Family | ID: MEM-9517 | Tel: 8884834606)
Secondary Caregivers
• Mary Ann (Rel: Family assistant | ID: MEM-8625 | Tel: 9946144439)
MEDICAL AUTHORITY
Consulting Doctor
Name: Dr. Anoop Thomas
Hospital: LLM Hospital, Kidangoor
Contact: 9870985643
CLINICAL NOTES:
"Needs continuous monitoring."
EMERGENCY ESCALATION
Level 1 Emergency Contact
Xavier Jacob | Neighbour | 9089765567
CONFIGURATION STATUS
Primary Caregiver Assigned
Secondary Caregiver Assigned
Emergency Contact Set

Alerts Page

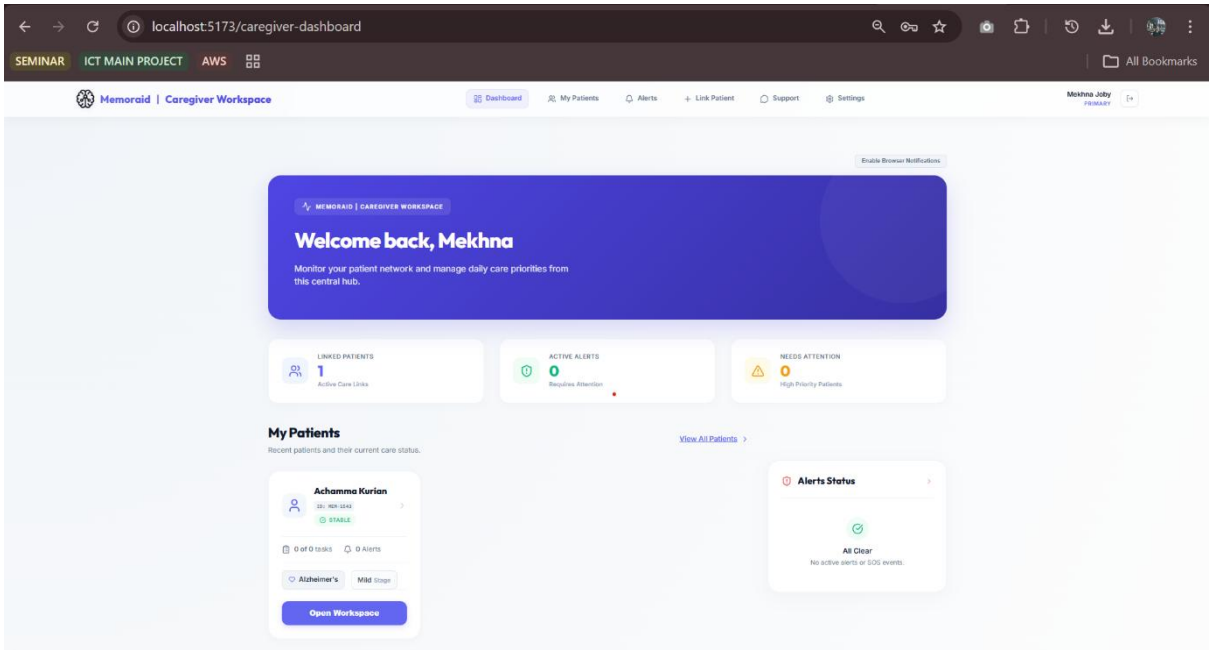

localhost:5173/admin/alerts
SEMINAR
ICT MAIN PROJECT
AWS
Memoraid | System Admin
Overview
Approvals
Users
Registry
Alerts
Inquiries
Logged in: System Admin
Logout
System Alert Monitor
Real-time oversight of all SOS triggers and missed routine escalations across the platform.
Active Alerts
Resolved
System History
SOS EMERGENCY
RESOLVED
Achamma Kurian: Patient requested immediate help via dashboard.
2/25/2026, 3:52:52 PM
Handled by: System Admin
Inspect Patient
MISSED ROUTINE ESCALATION
RESOLVED
Achamma Kurian: Routine 'Morning Medication: Tynol' scheduled at 09:30 AM was missed.
2/13/2026, 12:00:11 PM
Handled by: Mekhna Joby
Inspect Patient
MISSED ROUTINE ESCALATION
RESOLVED
Achamma Kurian: Acknowledged: Routine 'Lunch' scheduled at 12:45 PM was missed.
2/10/2026, 1:02:54 PM
Handled by: Mekhna Joby
Inspect Patient
MISSED ROUTINE ESCALATION
RESOLVED
Achamma Kurian: Acknowledged: Routine 'Daily Test' scheduled at 08:00 AM was missed.
2/10/2026, 12:52:59 PM
Handled by: Mekhna Joby
Inspect Patient

Inquiries and Complaints Page

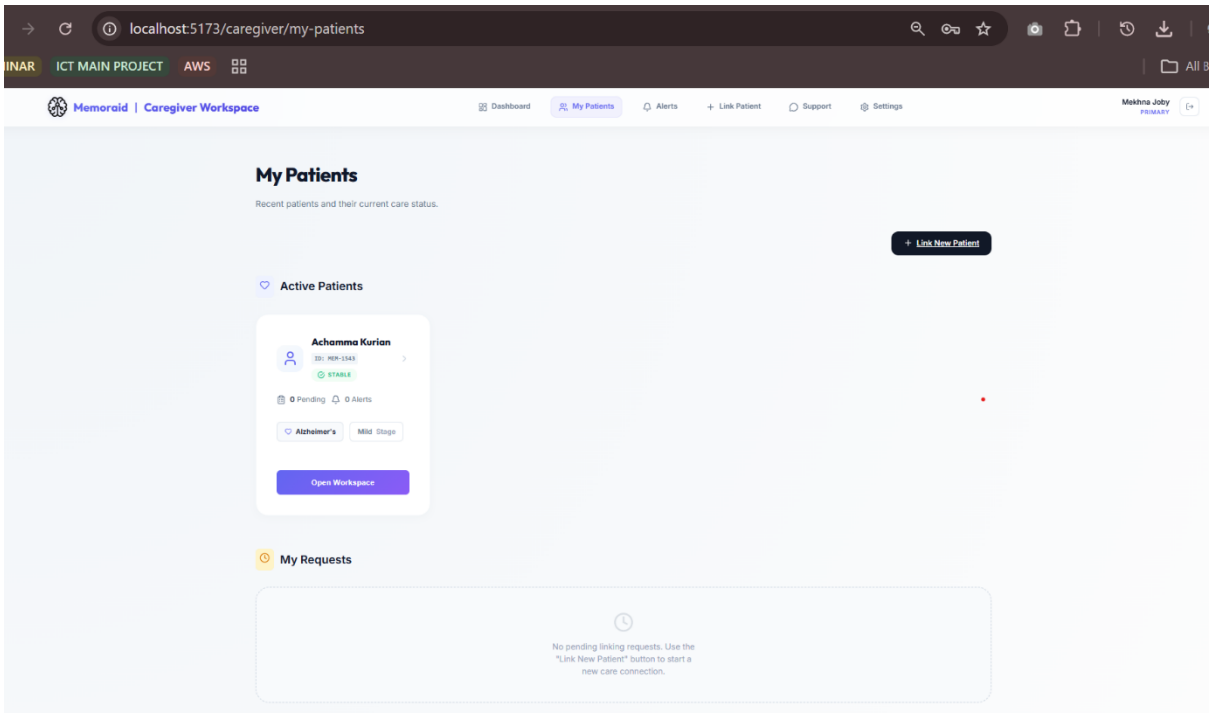


Caregiver Module

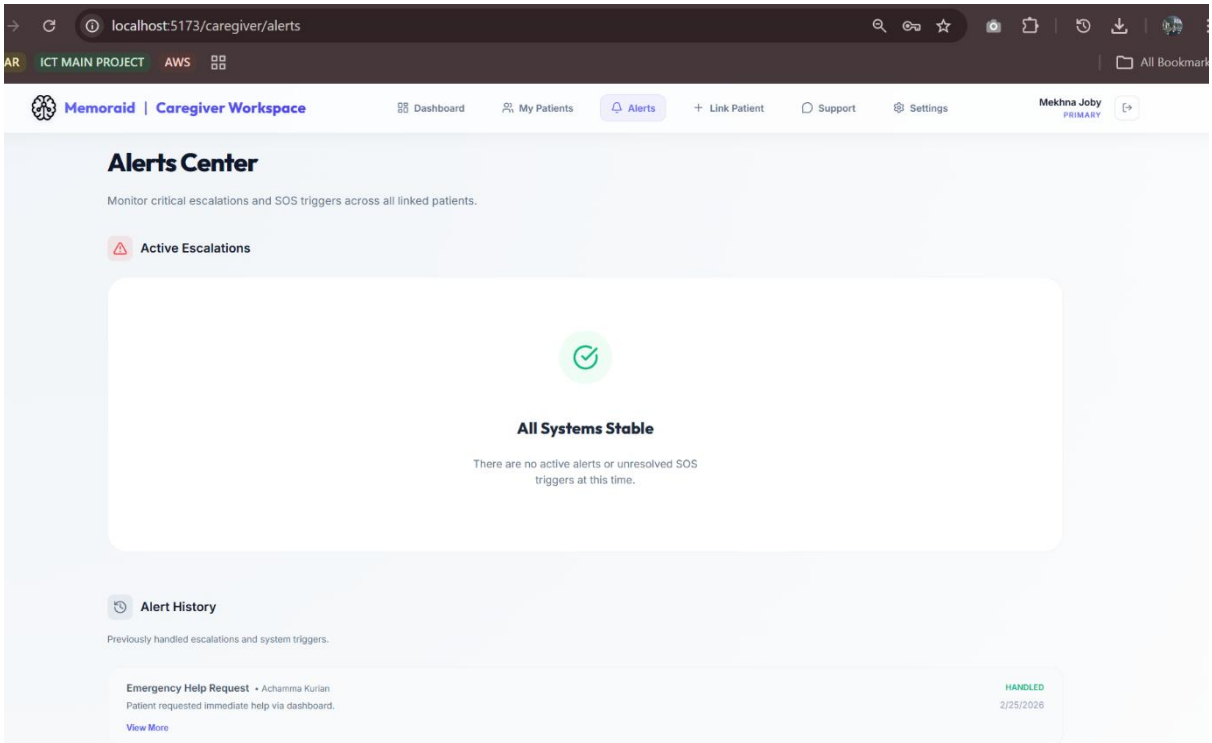
Dashboard Overview



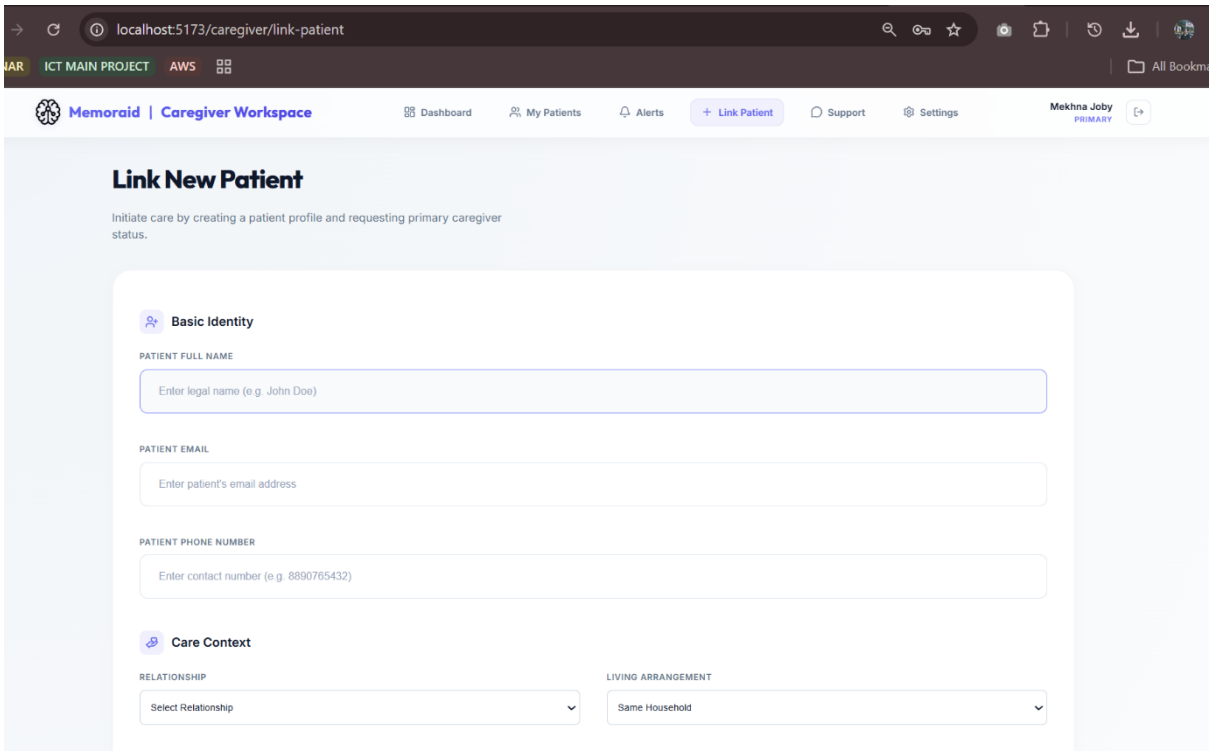
My Patients Page



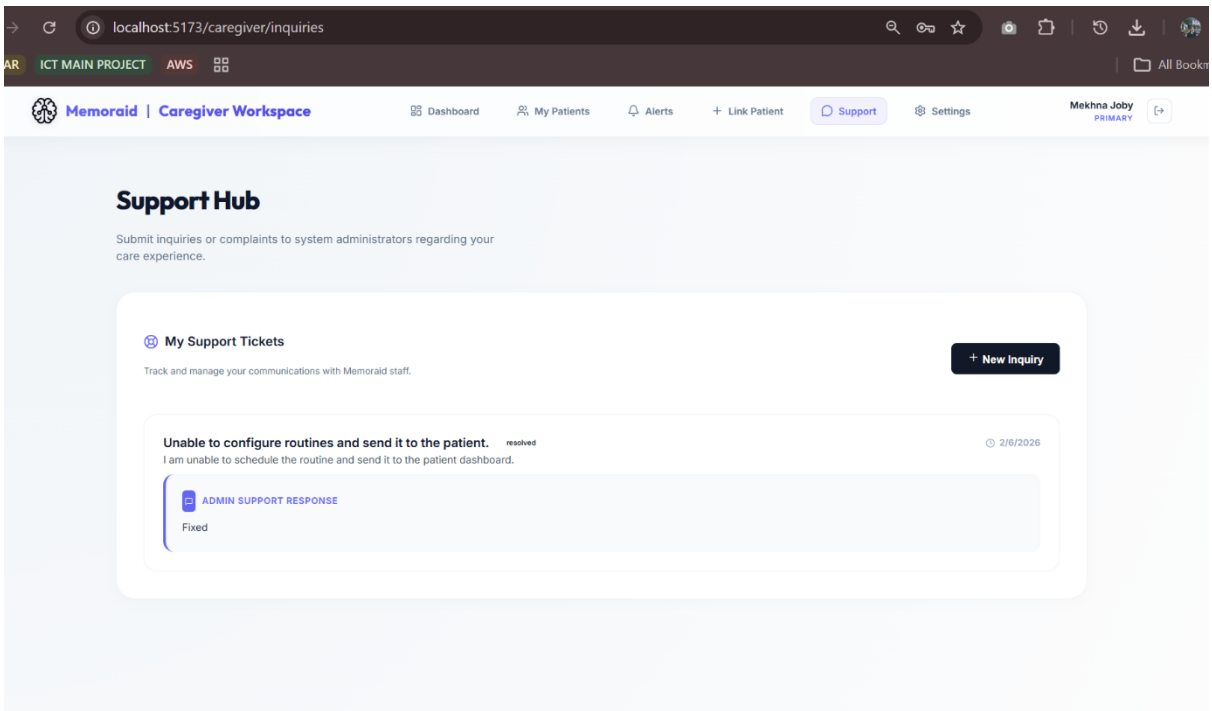
Alerts Page



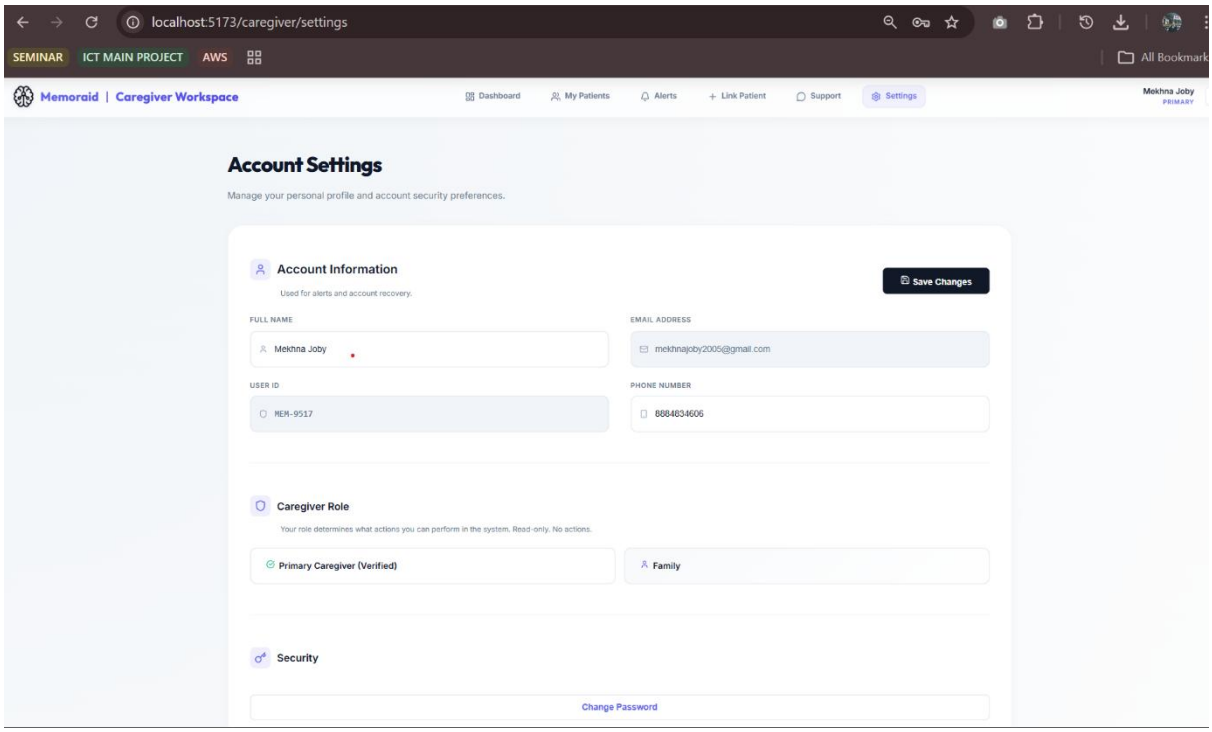
Link Patient Page



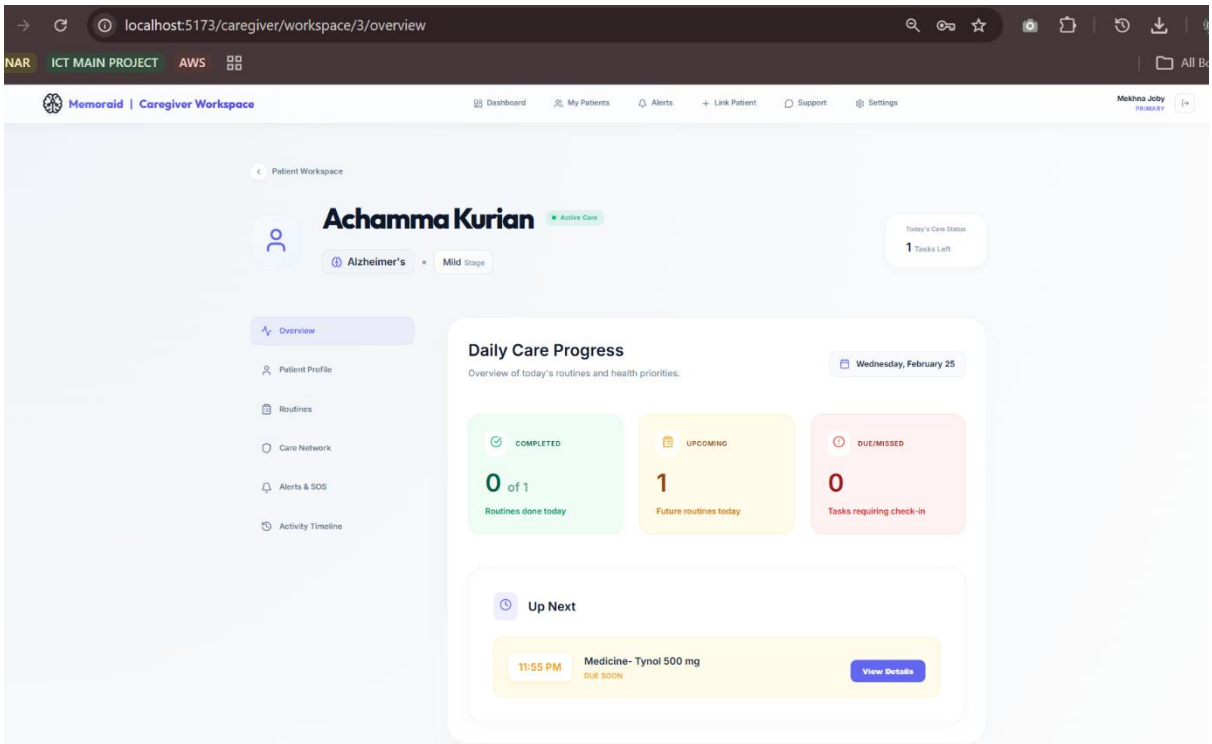
Support Center



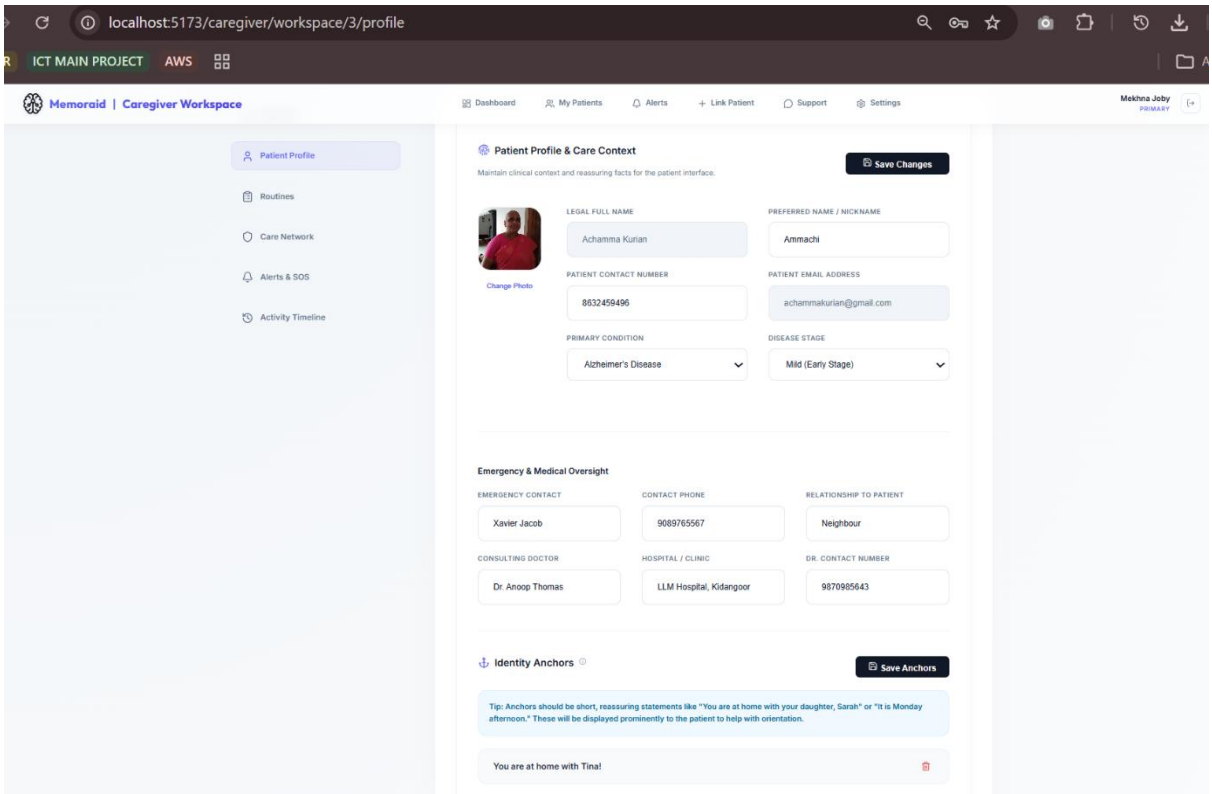
Settings Page



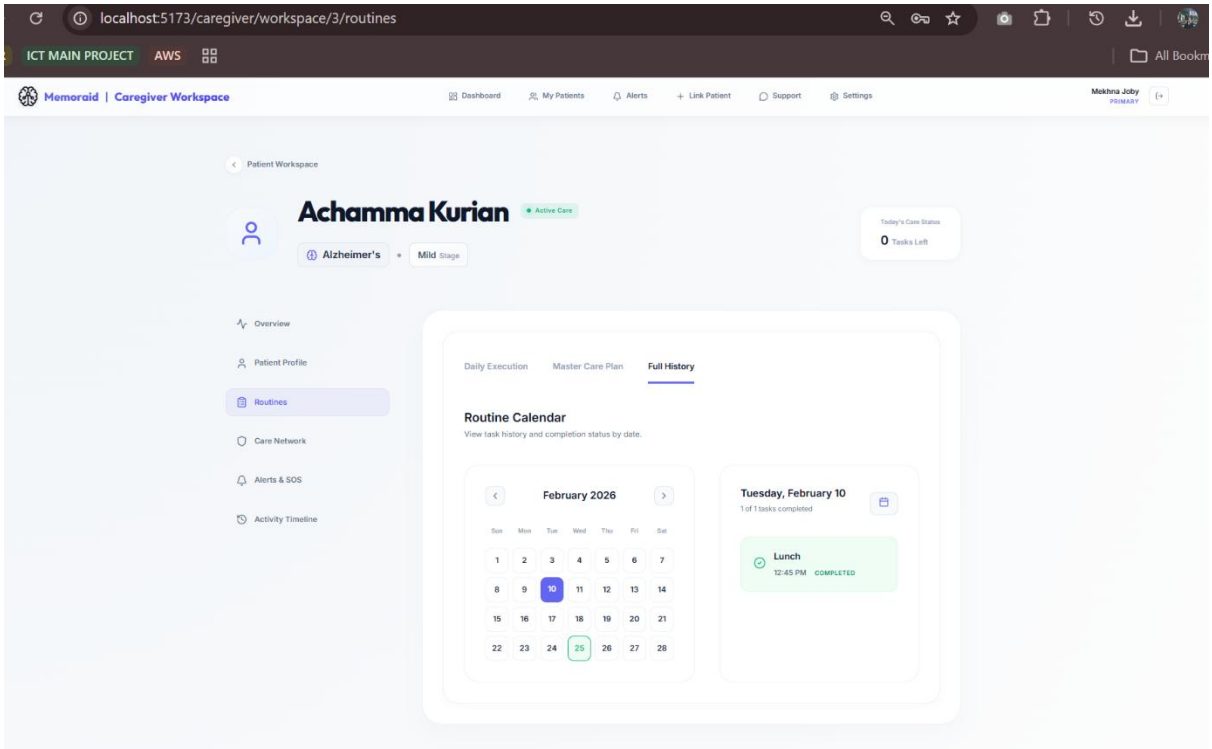
Patient Workspace Overview

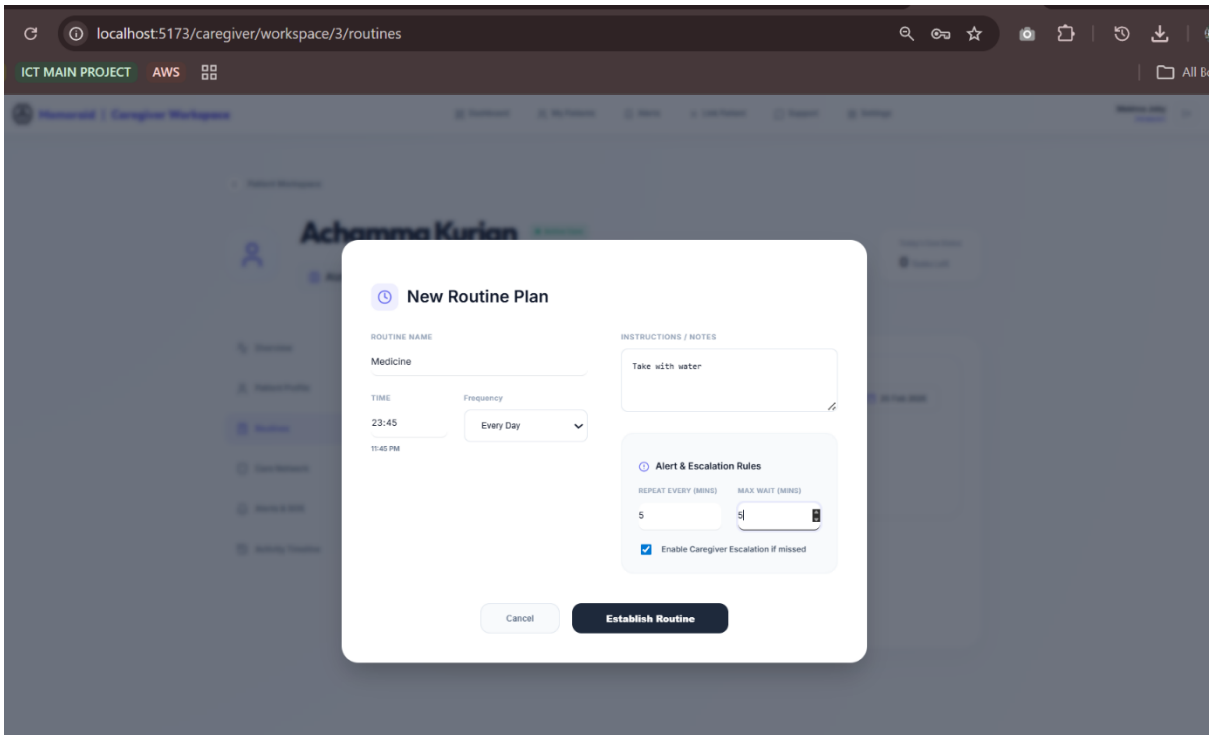


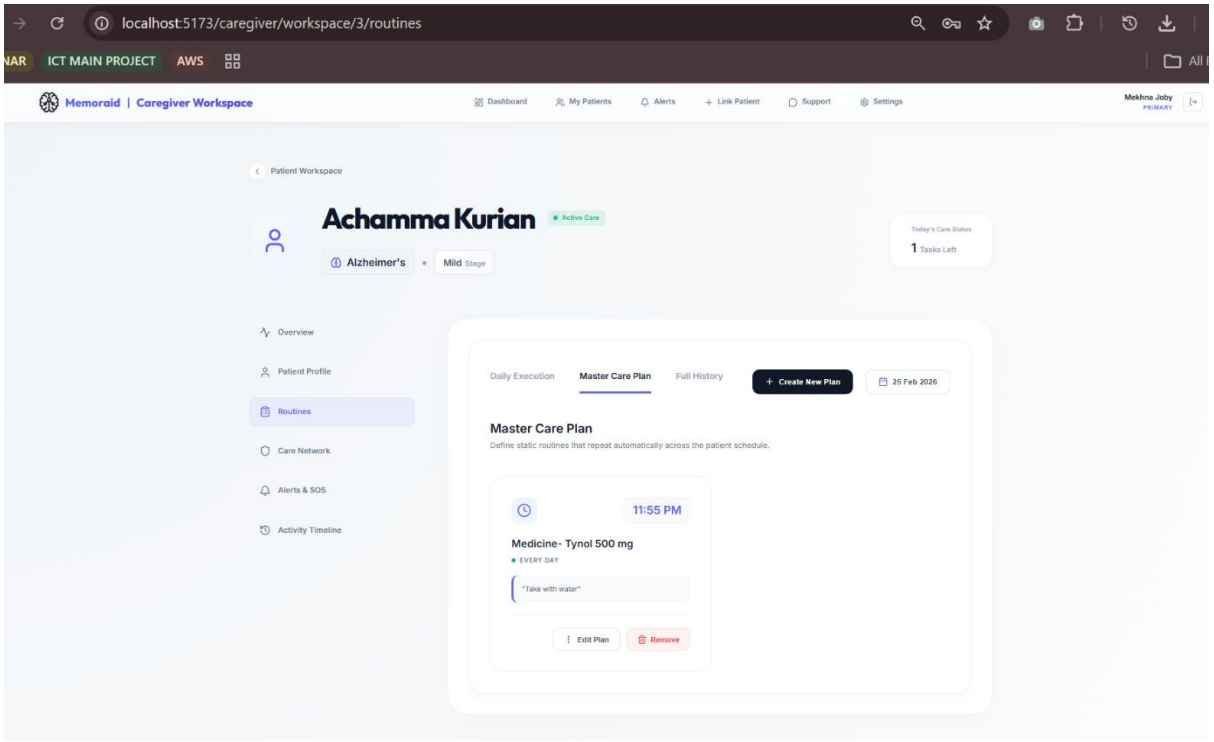
Patient’s Profile Page



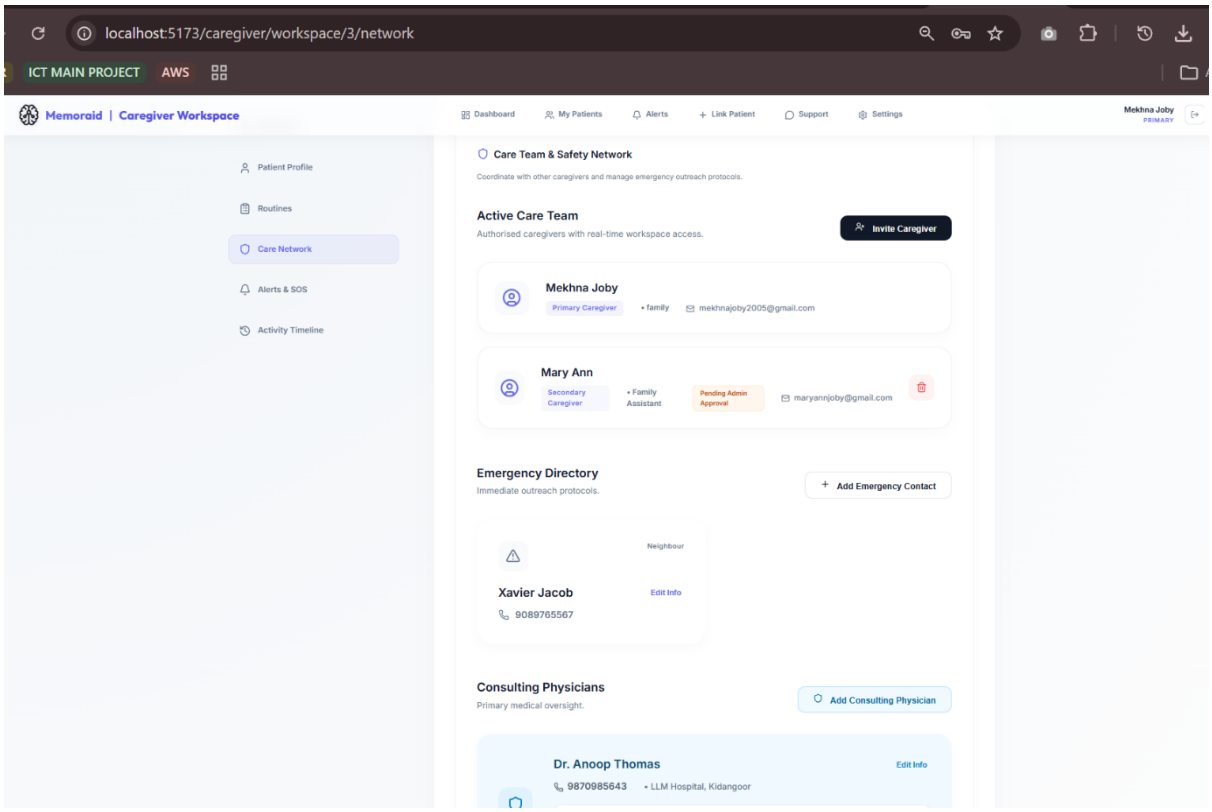
Routines Page



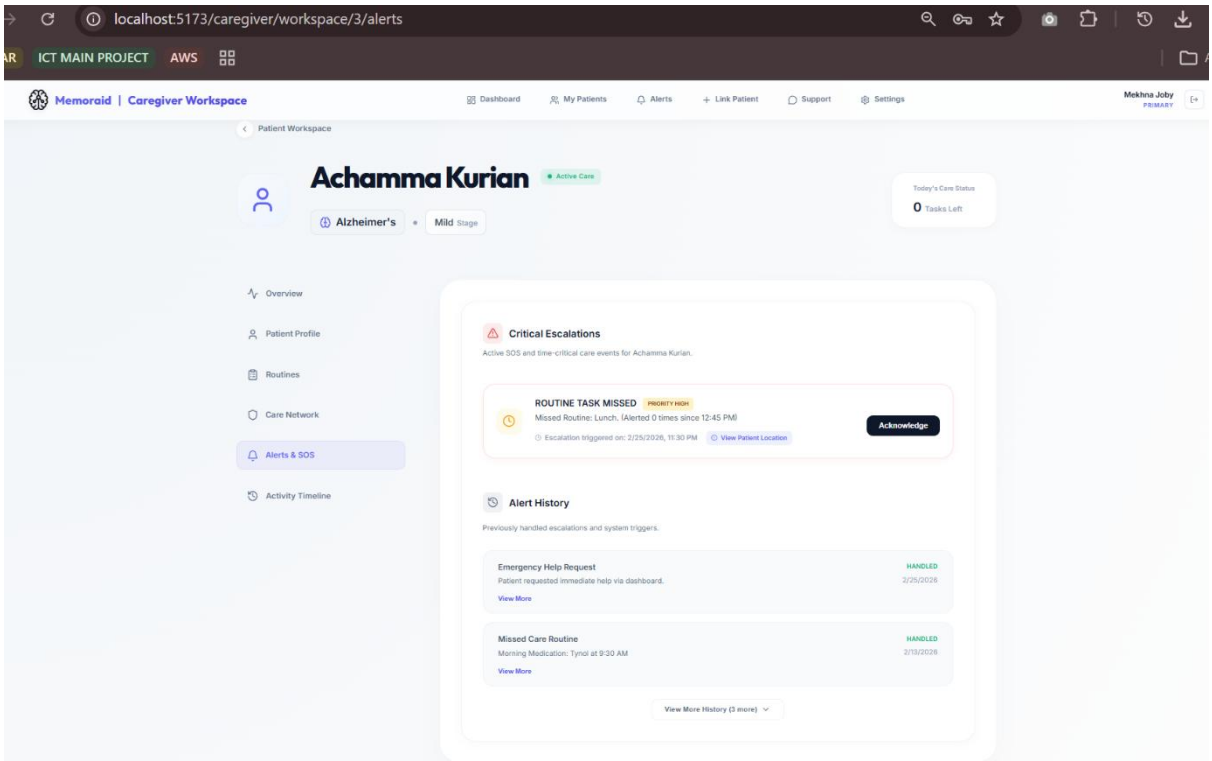

localhost:5173/caregiver/workspace/3/routines
ICT MAIN PROJECT
AWS
New Routine Plan
ROUTINE NAME
Medicine
INSTRUCTIONS / NOTES
Take with water
TIME
23:45
11:45 PM
Frequency
Every Day
Alert & Escalation Rules
REPEAT EVERY (MINS)
5
MAX WAIT (MINS)
5
Enable Caregiver Escalation if missed
Cancel
Establish Routine


localhost:5173/caregiver/workspace/3/routines
ICT MAIN PROJECT
AWS
Memoraid | Caregiver Workspace
Dashboard
My Patients
Alerts
Link Patient
Support
Settings
Mekhna Joby
PRIMARY
Patient Workspace
Achamma Kurian
Active Care
Alzheimer's
Mild Stage
Today's Care Status
1 Tasks Left
Overview
Patient Profile
Routines
Care Network
Alerts & SOS
Activity Timeline
Daily Execution
Master Care Plan
Full History
+ Create New Plan
25 Feb 2026
Master Care Plan
Define static routines that repeat automatically across the patient schedule.
11:55 PM
Medicine- Tynol 500 mg
EVERY DAY
"Take with water"
Edit Plan
Remove

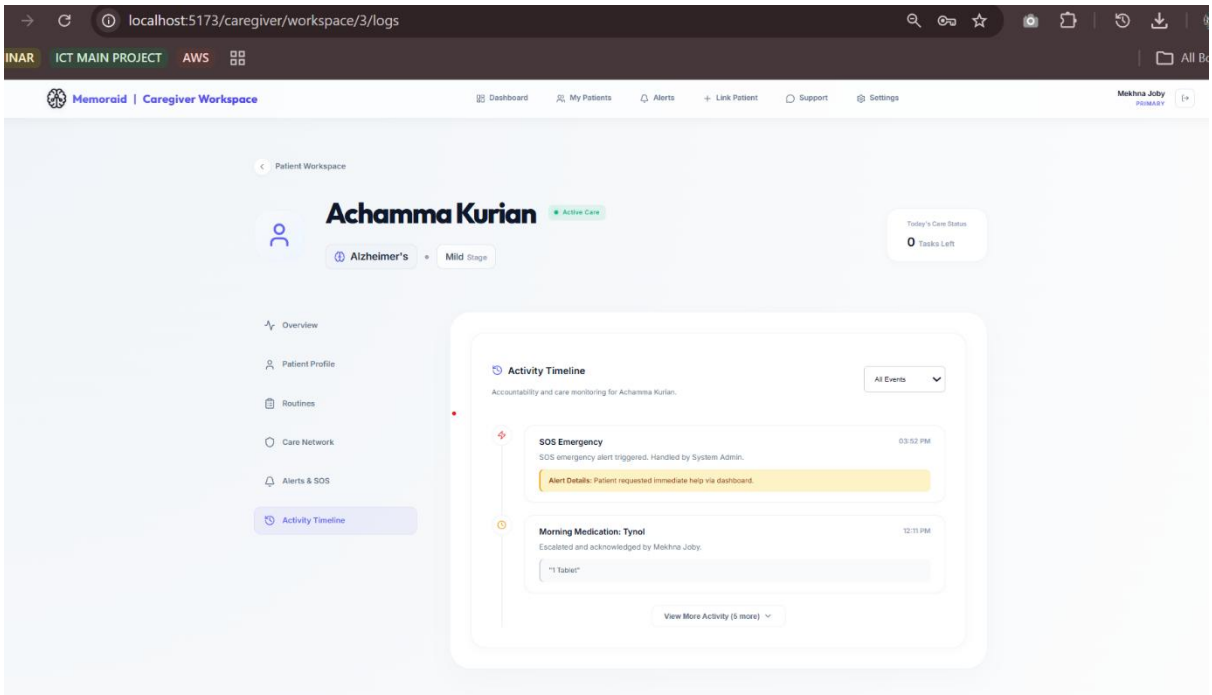
Care Network Page



Alerts & SOS Page

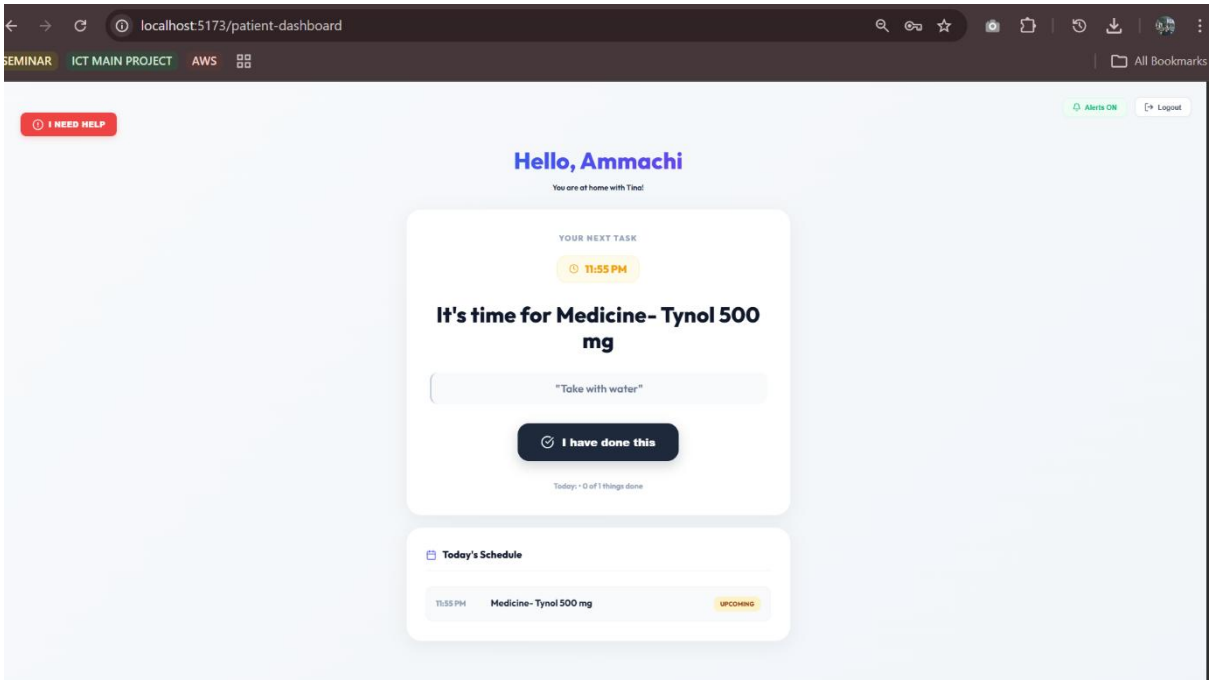
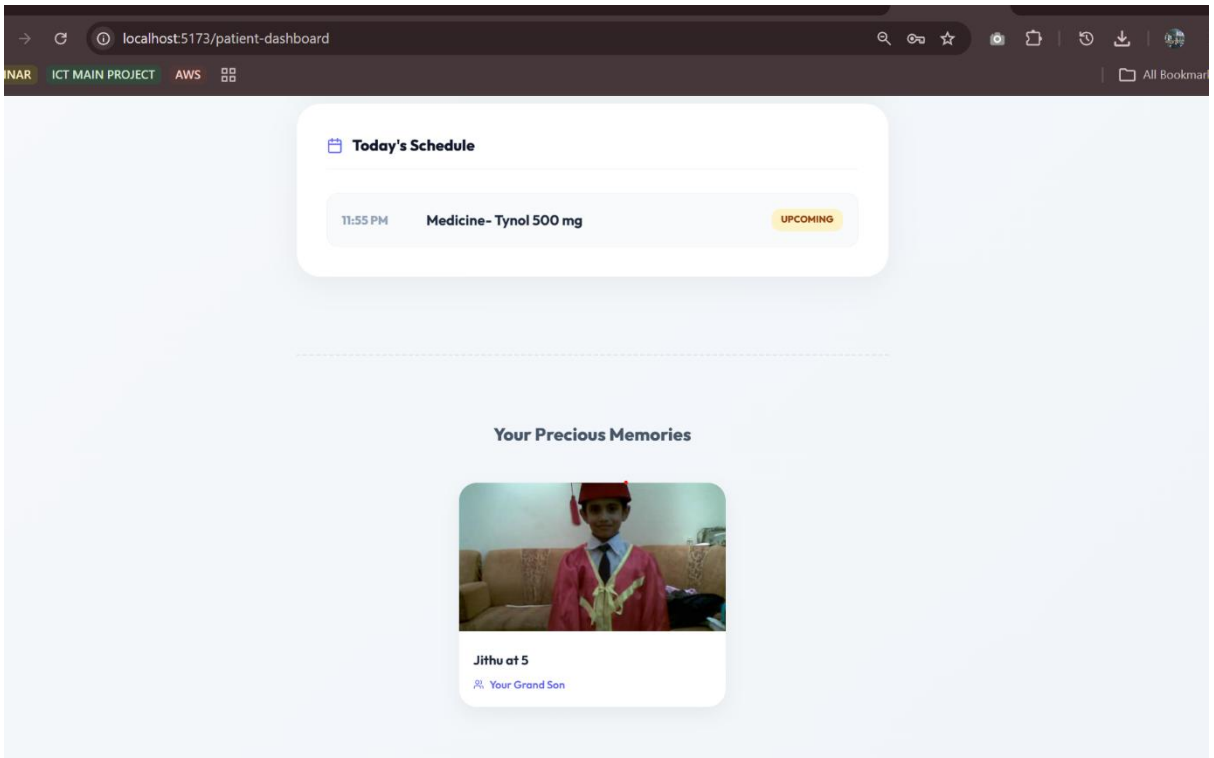


Activity Timeline Page



Patient Module

Dashboard Overview



Memory Gallery